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\\Research\Investigators\Frederick\CORPUS_CONSISTENT\rslh_ep3d_vaso
TA: 14 sec Coil Selection: Auto Voxel Size: 2.5×2.5×2.5 mm³ Acc:: None Rel. SNR: 1.00

Properties

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slab Group	1
Slabs	1
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
Slices per Slab	48
Phase Oversampling	0.0 %
Slice Oversampling	0.0 %
FOV Read	210 mm
FOV Phase	100.0 %
Slice Thickness	2.50 mm
TR	130.0 ms
Vol. TR	12.50064 s
TE 1	48.00 ms
Averages	1
Multi-echo Shots	1
AutoAlign	---

Contrast - Common

TR	130.0 ms
Vol. TR	12.50064 s
TE 1	48.00 ms
Multi-echo spacing	88.00 ms
MTC	Off
Magn. Preparation	Non-sel. IR
TI 1	3130.32 ms
TI 2	9370.32 ms
Flip Angle	13 deg
Fat-Water Contrast	Standard
Contrasts	1
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1

Contrast - Dynamic

Reordering	Linear

Resolution - Common

FOV Read	210 mm
FOV Phase	100.0 %
Slice Thickness	2.50 mm
Base Resolution	84
Phase Resolution	100 %
Slice Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	None
Phase Partial Fourier	Off
Slice Partial Fourier	Off
Asymmetric Echo	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	Off
Normalize	Off
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	1
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
Slices per Slab	48
Phase Oversampling	0.0 %
Slice Oversampling	0.0 %
FOV Read	210 mm
FOV Phase	100.0 %
Slice Thickness	2.50 mm
TR	130.0 ms
Vol. TR	12.50064 s
Multi-echo Shots	1

Geometry - AutoAlign

Slab Group	1
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
AutoAlign	---
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm

Geometry - AutoAlign

Initial Orientation	Transversal
Initial Rotation	0.00 deg

Geometry - Saturation

Saturation Mode	Standard

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Sum of Squares
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
A >> P	210 mm
R >> L	210 mm
F >> H	120 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Slab-sel.

System - Tx/Rx

Frequency 1H	123.249408 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Sequence - Part 1

Sequence Name	vso rslh6.0
Dimension	3D

Sequence - Part 1

Excitation	Slab-sel.
RF Pulse Type	Normal
Gradient Mode	Fast
Reordering	Linear
Bandwidth	1044 Hz/Px
Echo Spacing	1.04 ms
Asymmetric Echo	Off
Turbo Factor	48
Segmentation	1
EPI Factor	84

Sequence - Part 2

Introduction	On
RF Spoiling	On

Sequence - Special

RF duration	1080 us
RF time x BW	18
PAT ref. FA	5 deg
Fat sat. FA	110 deg
Variable FA	4
Invert PE	Off
Min. TE w/ PF	On
Expert ramping	Off
Trigger per shot	Off
Noise image	Off
Relax spoilers	Off
MT flip angle	170 deg
MT off-res.	2000 Hz
MT RF duration	12800 us
Phase Correction	per Series
EPI ramp factor	1.10
EPI ramp factor 2	1.10
Ramp sampling	1.00 1:max
Slab Scale	0 %
Modify Ice Config	On
G-factor map	Off
Mosaic DICOMs	Off
GRAPPA Regularization	5000 /10^6
G. spoil dephasing[1]	0 pi
G. spoil dephasing[2]	2 pi
G. spoil dephasing[3]	2 pi
Read polarity	Regular RO
VASO Pre-TI1 delay	0.00 ms
VASO Pre-TI2 delay	0 ms
VASO Pre-Inv delay	0 ms

Sequence - Assistant

SAR Assistant	Off

\\Research\Investigators\Frederick\CORPUS_CONSISTENT\eja_svs_steam_diff
TA: 3:39 min Coil Selection: Auto VoI: 20×20×20 mm³ Rel. SNR: 1.00

Properties

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
VoI A >> P	20 mm
VoI R >> L	20 mm
VoI F >> H	20 mm
TR	3000.0 ms
TE	135.00 ms
Averages	64

Contrast - Common

TR	3000.0 ms
TE	135.00 ms
TM	75.00 ms
Flip Angle	90 deg
Preparation Scans	4
Averages	64
VAPOR	Water Saturation
VAPOR suppr.	Water Suppr.
Water s. BW	135.00 Hz
Water s. Delta Pos.	0.00 ppm

Resolution - Common

Vector Size	2048
Normalize	Off

Geometry - Common

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
VoI A >> P	20 mm
VoI R >> L	20 mm
VoI F >> H	20 mm

Geometry - AutoAlign

AutoAlign	---
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Advanced
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off
Adj. Water Suppr.	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
A >> P	20 mm
R >> L	20 mm
F >> H	20 mm
Reset	Off

System - pTx

B1 Shim	TrueForm

System - Tx/Rx

Frequency 1H	123.249408 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	3000.0 ms

Sequence - Common

Sequence Name	steam

Sequence - Common

Preparation Scans	4
Delta Frequency	-1.70 ppm
Measurements	1
Phase Cycling	Auto
Save Uncombined	Off
Bandwidth	2000 Hz
Acquisition Duration	1024 ms
Remove Oversampling	On

Sequence - Nuclei

TX/RX Nucleus	1H
TX/RX Delta Frequency	0 Hz
TX Nucleus	None
TX Delta Frequency	0 Hz

Sequence - Special

RF pulse duration	2560 us
Spoiler max. amplitude	20.00 mT/m
Spoiler amp. ratio	1.50
Refocus grad. factor	1.00
OVS slab thickness	80.00 mm
OVS slab pos. offset	5.00 mm
OVS delta frequency	0.00 ppm
Spoiler duration	1300 us
Acq. window shift	200 us
Min. settling delay	300 us
Gradient ramp time	140 us
VAPOR flip angle	80 deg
VAPOR delay 8	28 ms
VAPOR delay 7	74 ms
OVS pulse duration	5120 us
OVS HS pulse N	1
OVS HS pulse R	40
OVS flip angle RO	90 deg
OVS flip angle PH	90 deg
OVS flip angle SL	90 deg
VAPOR delay 6	68 ms
VAPOR delay 5	106 ms
VAPOR delay 4	105 ms
VAPOR delay 3	146 ms
VAPOR delay 2	100 ms
VAPOR delay 1	150 ms
Diffusion weighting	Off
Enable OVS	On
Resolve averages	Off
Use expt. SAR calc	Off
Check RF clipping	Off
Send ref. scans	Off
Inversion pulse	Off
Symmetric RF pulses	Off
Invert SS grad. pol.	Off
Shift RO frequency	Off
Debug loop type	None

\\Research\Investigators\Frederick\CORPUS_CONSISTENT\ep2d_bold
TA: 12:42 min Coil Selection: Auto Voxel Size: 2.0×2.0×2.0 mm³ Acc:: 8 Rel. SNR: 1.00

Properties

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slice Group	1
Slices	64
Distance Factor	0 %
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FOV Read	200 mm
FOV Phase	100.0 %
Slice Thickness	2.0 mm
TR	627.0 ms
TE	30.00 ms
Averages	1
Concatenations	1
AutoAlign	---

Contrast - Common

TR	627.0 ms
TE	30.00 ms
MTC	Off
Flip Angle	60 deg
Fat-Water Contrast	Fat Saturation
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1200
Delay in TR	0.00 ms

Resolution - Common

FOV Read	200 mm
FOV Phase	100.0 %
Slice Thickness	2.0 mm
Base Resolution	100

Resolution - Common

Phase Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	SMS
Reference Scans	EPI/Separate
Acceleration Factor PE	1
SMS Factor	8
Phase Partial Fourier	7/8

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Hamming	Off
Distortion Correction	2D
Static Field Correction	Off
Normalize	Off

Geometry - Common

Slice Group	1
Slices	64
Distance Factor	0 %
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FOV Read	200 mm
FOV Phase	100.0 %
Slice Thickness	2.0 mm
TR	627.0 ms
Multi-Slice Mode	Interleaved
Series	Interleaved
Concatenations	1

Geometry - AutoAlign

Slice Group	1
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
AutoAlign	---
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

Geometry - Saturation

Special Saturation	None

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Sum of Squares
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Standard
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
A >> P	200 mm
R >> L	200 mm
F >> H	128 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Standard

System - Tx/Rx

Frequency 1H	123.249408 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	627.0 ms
Log Signals	Off
Concatenations	1

BOLD

GLM Statistics	Off
Ignore Meas. at Start	0
Ignore After Transition	0

BOLD

Model Transition States	Off
Temp. Highpass Filter	Off
Threshold	4.00
Paradigm Size	40
Meas[1]	Baseline
Meas[2]	Baseline
Meas[3]	Baseline
Meas[4]	Baseline
Meas[5]	Baseline
Meas[6]	Baseline
Meas[7]	Baseline
Meas[8]	Baseline
Meas[9]	Baseline
Meas[10]	Baseline
Meas[11]	Baseline
Meas[12]	Baseline
Meas[13]	Baseline
Meas[14]	Baseline
Meas[15]	Baseline
Meas[16]	Baseline
Meas[17]	Baseline
Meas[18]	Baseline
Meas[19]	Baseline
Meas[20]	Baseline
Meas[21]	Active
Meas[22]	Active
Meas[23]	Active
Meas[24]	Active
Meas[25]	Active
Meas[26]	Active
Meas[27]	Active
Meas[28]	Active
Meas[29]	Active
Meas[30]	Active
Meas[31]	Active
Meas[32]	Active
Meas[33]	Active
Meas[34]	Active
Meas[35]	Active
Meas[36]	Active
Meas[37]	Active
Meas[38]	Active
Meas[39]	Active
Meas[40]	Active
Motion Correction	Off
Spatial Filter	Off
Measurements	1200
Delay in TR	0.00 ms

Sequence - Part 1

Sequence Name	epfid
Excitation	Standard
RF Pulse Type	Normal
Gradient Mode	Performance

Sequence - Part 1

Bandwidth	2174 Hz/Px
Echo Spacing	0.62 ms
Free Echo Spacing	Off
EPI Factor	100

Sequence - Part 2

Introduction	On

Sequence - Assistant

SAR Assistant	Off

\\Research\Investigators\Frederick\CORPUS_CONSISTENT\ep2d_bold_MGH

TA: 6:21 min Coil Selection: Auto Voxel Size: 3.0×3.0×3.0 mm³ Acc:: None Rel. SNR: 1.00

Properties

Start measurement without further preparation	On
Wait for User to Start	On
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	On
Auto Close Inline Display	On
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	On
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slice Group	1
Slices	47
Distance Factor	0 %
Position	R3.0 A3.0 H0.0 mm
Orientation	T > C-12.5
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FOV Read	216 mm
FOV Phase	100.0 %
Slice Thickness	3.0 mm
TR	3070.0 ms
TE	30.00 ms
Averages	1
Concatenations	1
AutoAlign	Head > Basis

Contrast - Common

TR	3070.0 ms
TE	30.00 ms
MTC	Off
Flip Angle	85 deg
Fat-Water Contrast	Fat Saturation
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	124
Delay in TR	0.00 ms

Resolution - Common

FOV Read	216 mm
FOV Phase	100.0 %
Slice Thickness	3.0 mm
Base Resolution	72
Phase Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	None
Phase Partial Fourier	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Hamming	Off
Distortion Correction	Off
Static Field Correction	Off
Normalize	Off

Geometry - Common

Slice Group	1
Slices	47
Distance Factor	0 %
Position	R3.0 A3.0 H0.0 mm
Orientation	T > C-12.5
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FOV Read	216 mm
FOV Phase	100.0 %
Slice Thickness	3.0 mm
TR	3070.0 ms
Multi-Slice Mode	Interleaved
Series	Interleaved
Concatenations	1

Geometry - AutoAlign

Slice Group	1
Position	R3.0 A3.0 H0.0 mm
Orientation	T > C-12.5
Phase Encoding Dir.	A >> P
AutoAlign	Head > Basis
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

Geometry - Saturation

Special Saturation	None

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm

Geometry - Tim Planning Suite

Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Sum of Squares
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Standard
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	R3.0 A3.0 H0.0 mm
Orientation	T > C-12.5
Rotation	0.00 deg
A >> P	216 mm
R >> L	216 mm
F >> H	141 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Standard

System - Tx/Rx

Frequency 1H	123.249408 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	3070.0 ms
Log Signals	Off
Concatenations	1

BOLD

GLM Statistics	Off
Ignore Meas. at Start	0
Ignore After Transition	0
Model Transition States	Off
Temp. Highpass Filter	Off
Threshold	4.00
Paradigm Size	3
Meas[1]	Baseline
Meas[2]	Baseline
Meas[3]	Active
Motion Correction	Off
Spatial Filter	Off
Measurements	124
Delay in TR	0.00 ms

Sequence - Part 1

Sequence Name	epfid
Excitation	Standard
RF Pulse Type	Normal
Gradient Mode	Fast
Bandwidth	2240 Hz/Px
Echo Spacing	0.51 ms
Free Echo Spacing	Off
EPI Factor	72

Sequence - Part 2

Introduction	Off

Sequence - Special

Imaging Dummy TRs	0
SMS ACS Dummy TRs	-1
FLEET Dummy Pulses	0
CC Mode	Direct adj coil
RF Clip	0
VERSE Factor	1.00
ACS mode	Standard
Kernel Size	3x3
Wait for TCP/IP Trigger	Off
Reverse Phase Encoding	Off

Sequence - Assistant

SAR Assistant	Off

\\Research\Investigators\Frederick\CORPUS_CONSISTENT\ep2d_bold_mgh
TA: 8 sec Coil Selection: Auto Voxel Size: 3.9×3.9×5.0 mm³ Acc:: None Rel. SNR: 1.00

Properties

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slice Group	1
Slices	1
Distance Factor	50 %
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FOV Read	500 mm
FOV Phase	100.0 %
Slice Thickness	5.0 mm
TR	223.0 ms
TE	98.00 ms
Averages	1
Concatenations	1
AutoAlign	---

Contrast - Common

TR	223.0 ms
TE	98.00 ms
MTC	Off
Flip Angle	90 deg
Fat-Water Contrast	Fat Saturation
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	20
Delay in TR	0.00 ms

Resolution - Common

FOV Read	500 mm
FOV Phase	100.0 %
Slice Thickness	5.0 mm
Base Resolution	128

Resolution - Common

Phase Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	None
Phase Partial Fourier	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Hamming	Off
Distortion Correction	2D
Static Field Correction	Off
Normalize	Off

Geometry - Common

Slice Group	1
Slices	1
Distance Factor	50 %
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FOV Read	500 mm
FOV Phase	100.0 %
Slice Thickness	5.0 mm
TR	223.0 ms
Multi-Slice Mode	Interleaved
Series	Interleaved
Concatenations	1

Geometry - AutoAlign

Slice Group	1
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
AutoAlign	---
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

Geometry - Saturation

Special Saturation	None

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm

Geometry - Tim Planning Suite

Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Sum of Squares
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Standard
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
A >> P	500 mm
R >> L	500 mm
F >> H	5 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Standard

System - Tx/Rx

Frequency 1H	123.249408 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	223.0 ms
Log Signals	Off
Concatenations	1

BOLD

GLM Statistics	On
Ignore Meas. at Start	0
Ignore After Transition	0
Model Transition States	On
Temp. Highpass Filter	On

BOLD

Threshold	4.00
Paradigm Size	20
Meas[1]	Baseline
Meas[2]	Baseline
Meas[3]	Baseline
Meas[4]	Baseline
Meas[5]	Baseline
Meas[6]	Baseline
Meas[7]	Baseline
Meas[8]	Baseline
Meas[9]	Baseline
Meas[10]	Baseline
Meas[11]	Active
Meas[12]	Active
Meas[13]	Active
Meas[14]	Active
Meas[15]	Active
Meas[16]	Active
Meas[17]	Active
Meas[18]	Active
Meas[19]	Active
Meas[20]	Active
Motion Correction	On
Spatial Filter	Off
Measurements	20
Delay in TR	0.00 ms

Sequence - Part 1

Sequence Name	epfid
Excitation	Standard
RF Pulse Type	Normal
Gradient Mode	Fast
Bandwidth	752 Hz/Px
Echo Spacing	1.38 ms
Free Echo Spacing	Off
EPI Factor	128

Sequence - Part 2

Introduction	Off

Sequence - Special

Imaging Dummy TRs	-1
SMS ACS Dummy TRs	-1
FLEET Dummy Pulses	0
CC Mode	Direct adj coil
RF Clip	0
VERSE Factor	1.00
ACS mode	Standard
Kernel Size	3x3
Wait for TCP/IP Trigger	Off
Reverse Phase Encoding	Off

Sequence - Assistant

SAR Assistant	Off

\\Research\Investigators\Frederick\CORPUS_CONSISTENT\ep2d_diff
TA: 36 sec Coil Selection: Auto Voxel Size: 3.3×3.3×1.5 mm³ Acc:: None Rel. SNR: 1.00

Properties

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slice Group	1
Slices	28
Distance Factor	0 %
Position	Isocenter
Orientation	T > C-16.0
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FOV Read	210 mm
FOV Phase	100.0 %
Slice Thickness	1.5 mm
TR	4500.0 ms
TE	60.00 ms
Concatenations	1
AutoAlign	Head > Brain

Contrast - Common

TR	4500.0 ms
TE	60.00 ms
MTC	Off
Magn. Preparation	None
Fat-Water Contrast	Fat Saturation
Fat Saturation	Weak
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Multiple Series	Off
Delay in TR	0.00 ms

Resolution - Common

FOV Read	210 mm
FOV Phase	100.0 %
Slice Thickness	1.5 mm
Base Resolution	64

Resolution - Common

Phase Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	None
Deep Resolve	Off
Phase Partial Fourier	6/8

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	Off
Static Field Correction	Off
Normalize	Off

Geometry - Common

Slice Group	1
Slices	28
Distance Factor	0 %
Position	Isocenter
Orientation	T > C-16.0
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FOV Read	210 mm
FOV Phase	100.0 %
Slice Thickness	1.5 mm
TR	4500.0 ms
Multi-Slice Mode	Interleaved
Series	Interleaved
Concatenations	1

Geometry - AutoAlign

Slice Group	1
Position	Isocenter
Orientation	T > C-16.0
Phase Encoding Dir.	A >> P
AutoAlign	Head > Brain
Initial Position	L0.0 P0.0 H6.0
L	0.0 mm
P	0.0 mm
H	6.0 mm
Initial Orientation	T > C
T > C	-16.00
> S	0.00
Initial Rotation	0.00 deg

Geometry - Navigator

Geometry - Saturation

Special Saturation	None

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	6 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	ACS All but spine
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Performance
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Standard
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	Isocenter
Orientation	T > C-16.0
Rotation	0.00 deg
A >> P	210 mm
R >> L	210 mm
F >> H	42 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Standard

System - Tx/Rx

Frequency 1H	123.249408 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	4500.0 ms
Concatenations	1

Physio - PACE

Resp. Control	Off
Concatenations	1

Diff

Diffusion Mode	MDDW
Diff. Directions	6
Diffusion Scheme	Bipolar
Diff. Weightings	2
b-value 1	0 s/mm²
b-value 2	1500 s/mm²
Averages 1	1
Averages 2	1
Dynamic Field Correction	Off
Invert Gray Scale	Off
Diff. Weighted Images	On
Trace Weighted Images	Off
Tensor	Off
FA Maps	Off
ADC Maps	On
Exponential ADC Maps	Off
ADC Noise Threshold	40
Calculated Image	Off

Sequence - Part 1

Sequence Name	epse
Excitation	Standard
RF Pulse Type	Normal
Gradient Mode	Performance
Bandwidth	1776 Hz/Px
Echo Spacing	0.63 ms
Free Echo Spacing	Off
Optimization	Min. TE
EPI Factor	64

Sequence - Part 2

Introduction	Off
Phase Correction	Internal

Sequence - Assistant

SAR Assistant	Off
Optimization	Min. TE

\\Research\Investigators\Frederick\CORPUS_CONSISTENT\ep2d_diff_mgh
TA: 1 sec Coil Selection: Auto Voxel Size: 3.9×3.9×5.0 mm³ Acc:: None Rel. SNR: 1.00

Properties

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slice Group	1
Slices	1
Distance Factor	50 %
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FOV Read	500 mm
FOV Phase	100.0 %
Slice Thickness	5.0 mm
TR	500.0 ms
TE	200.00 ms
Concatenations	1
AutoAlign	---

Contrast - Common

TR	500.0 ms
TE	200.00 ms
MTC	Off
Magn. Preparation	None
Flip Angle 1	90 deg
Flip Angle 2	180 deg
Fat-Water Contrast	Fat Saturation
Fat Saturation	Weak
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Delay in TR	0.00 ms

Resolution - Common

FOV Read	500 mm
FOV Phase	100.0 %
Slice Thickness	5.0 mm

Resolution - Common

Base Resolution	128
Phase Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	None
Phase Partial Fourier	6/8

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	2D
Static Field Correction	Off
Normalize	Off

Geometry - Common

Slice Group	1
Slices	1
Distance Factor	50 %
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FOV Read	500 mm
FOV Phase	100.0 %
Slice Thickness	5.0 mm
TR	500.0 ms
Multi-Slice Mode	Interleaved
Series	Interleaved
Concatenations	1

Geometry - AutoAlign

Slice Group	1
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
AutoAlign	---
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

Geometry - Navigator

Geometry - Saturation

Special Saturation	None

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Standard
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
A >> P	500 mm
R >> L	500 mm
F >> H	5 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Standard

System - Tx/Rx

Frequency 1H	123.249408 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	500.0 ms
Concatenations	1

Physio - PACE

Resp. Control	Off
Concatenations	1

Diff

Diffusion Mode	Read
Diff. Directions	1
Diffusion Scheme	Bipolar
Diff. Weightings	1
b-value	0 s/mm²
Averages	1
Dynamic Field Correction	Off
Invert Gray Scale	Off
Diff. Weighted Images	On
Trace Weighted Images	Off
Tensor	Off
FA Maps	Off
ADC Maps	Off
Exponential ADC Maps	Off
Calculated Image	Off

Sequence - Part 1

Sequence Name	epse
Excitation	Standard
RF Pulse Type	Normal
Gradient Mode	Fast
Bandwidth	752 Hz/Px
Echo Spacing	1.38 ms
Free Echo Spacing	Off
Optimization	None
EPI Factor	128

Sequence - Part 2

Introduction	Off
Phase Correction	Internal

Sequence - Special

Imaging Dummy TRs	-1
SMS ACS Dummy TRs	-1
FLEET Dummy Pulses	0
RF Clip	0
VERSE Factor	1.00
ACS mode	Standard
Kernel Size	3x3
Wait for TCP/IP Trigger	Off
Reverse Phase Encoding	Off

Sequence - Assistant

SAR Assistant	Off
Optimization	None

\\Research\Investigators\Frederick\CORPUS_CONSISTENT\ep2d_fid

TA: 6 sec Coil Selection: Auto Voxel Size: 0.9×0.9×5.0 mm³ Acc:: None Rel. SNR: 1.00

Properties

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	On
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slice Group	1
Slices	25
Distance Factor	20 %
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FOV Read	220 mm
FOV Phase	100.0 %
Slice Thickness	5.0 mm
TR	6120.0 ms
TE	30.00 ms
Averages	1
Concatenations	1
AutoAlign	Head > Brain

Contrast - Common

TR	6120.0 ms
TE	30.00 ms
MTC	Off
Flip Angle	90 deg
Fat-Water Contrast	Fat Saturation
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Multiple Series	Off
Delay in TR	0.00 ms

Resolution - Common

FOV Read	220 mm
FOV Phase	100.0 %
Slice Thickness	5.0 mm

Resolution - Common

Base Resolution	128
Phase Resolution	100 %
Interpolation	On

Resolution - Acceleration

Acceleration Mode	None
Phase Partial Fourier	6/8

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Hamming	Off
Distortion Correction	2D
Static Field Correction	Off
Normalize	Prescan

Geometry - Common

Slice Group	1
Slices	25
Distance Factor	20 %
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FOV Read	220 mm
FOV Phase	100.0 %
Slice Thickness	5.0 mm
TR	6120.0 ms
Multi-Slice Mode	Interleaved
Series	Interleaved
Concatenations	1

Geometry - AutoAlign

Slice Group	1
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
AutoAlign	Head > Brain
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.01 deg

Geometry - Saturation

Special Saturation	None

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off

Geometry - Tim Planning Suite

Table Position	0 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Sum of Squares
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Standard
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.01 deg
A >> P	220 mm
R >> L	220 mm
F >> H	149 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Standard

System - Tx/Rx

Frequency 1H	123.249408 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	6120.0 ms
Concatenations	1

Perf

Measurements	1

Sequence - Part 1

Sequence Name	epfid
Excitation	Standard

Sequence - Part 1

RF Pulse Type	Normal
Gradient Mode	Fast
Bandwidth	1502 Hz/Px
Echo Spacing	0.77 ms
Free Echo Spacing	On
EPI Factor	128

Sequence - Part 2

Introduction	Off

Sequence - Assistant

SAR Assistant	Off

\\Research\Investigators\Frederick\CORPUS_CONSISTENT\ep2d_pace
TA: 6:44 min Coil Selection: Auto Voxel Size: 3.4×3.4×4.0 mm³ Acc:: None Rel. SNR: 1.00

Properties

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slice Group	1
Slices	28
Distance Factor	25 %
Position	R4.1 A8.1 F33.2 mm
Orientation	Transversal
Phase Encoding Dir.	P >> A
Phase Oversampling	0 %
FOV Read	220 mm
FOV Phase	100.0 %
Slice Thickness	4.0 mm
TR	2000.0 ms
TE	30.00 ms
Averages	1
Concatenations	1
AutoAlign	---

Contrast - Common

TR	2000.0 ms
TE	30.00 ms
MTC	Off
Flip Angle	90 deg
Fat-Water Contrast	Fat Saturation
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	200
Delay in TR	0.00 ms

Resolution - Common

FOV Read	220 mm
FOV Phase	100.0 %
Slice Thickness	4.0 mm
Base Resolution	64

Resolution - Common

Phase Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	None
Phase Partial Fourier	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Hamming	Off
Distortion Correction	Off
Static Field Correction	Off
Normalize	Off

Geometry - Common

Slice Group	1
Slices	28
Distance Factor	25 %
Position	R4.1 A8.1 F33.2 mm
Orientation	Transversal
Phase Encoding Dir.	P >> A
Phase Oversampling	0 %
FOV Read	220 mm
FOV Phase	100.0 %
Slice Thickness	4.0 mm
TR	2000.0 ms
Multi-Slice Mode	Interleaved
Series	Interleaved
Concatenations	1

Geometry - AutoAlign

Slice Group	1
Position	R4.1 A8.1 F33.2 mm
Orientation	Transversal
Phase Encoding Dir.	P >> A
AutoAlign	---
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

Geometry - Saturation

Special Saturation	None

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm

Geometry - Tim Planning Suite

Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Sum of Squares
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Standard
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	R4.1 A8.1 F33.2 mm
Orientation	Transversal
Rotation	180.00 deg
A >> P	220 mm
R >> L	220 mm
F >> H	139 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Standard

System - Tx/Rx

Frequency 1H	123.249408 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	2000.0 ms
Log Signals	Off
Concatenations	1

BOLD

GLM Statistics	Off
Ignore Meas. at Start	0
Ignore After Transition	0
Model Transition States	On
Temp. Highpass Filter	On

BOLD

Threshold	4.00
Paradigm Size	20
Meas[1]	Baseline
Meas[2]	Baseline
Meas[3]	Baseline
Meas[4]	Baseline
Meas[5]	Baseline
Meas[6]	Baseline
Meas[7]	Baseline
Meas[8]	Baseline
Meas[9]	Baseline
Meas[10]	Baseline
Meas[11]	Active
Meas[12]	Active
Meas[13]	Active
Meas[14]	Active
Meas[15]	Active
Meas[16]	Active
Meas[17]	Active
Meas[18]	Active
Meas[19]	Active
Meas[20]	Active
Motion Correction	On
Spatial Filter	Off
Measurements	200
Delay in TR	0.00 ms

Sequence - Part 1

Sequence Name	epfid
Excitation	Standard
RF Pulse Type	Normal
Gradient Mode	Fast
Bandwidth	1662 Hz/Px
Echo Spacing	0.67 ms
Free Echo Spacing	Off
EPI Factor	64

Sequence - Part 2

Introduction	Off

Sequence - Assistant

SAR Assistant	Off

\\Research\Investigators\Frederick\CORPUS_CONSISTENT\ep2d_se
TA: 29 sec Coil Selection: Auto Voxel Size: 2.7×2.7×2.5 mm³ Acc:: 2 Rel. SNR: 1.00

Properties

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	On
Graphic segment	2nd Segment
Inline Movie	Off

Routine

Slice Group	1
Slices	60
Distance Factor	0 %
Position	R2.8 A11.0 H5.9 mm
Orientation	Transversal
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FOV Read	235 mm
FOV Phase	100.0 %
Slice Thickness	2.5 mm
TR	7080.0 ms
TE	75.00 ms
Averages	1
Concatenations	3
AutoAlign	---

Contrast - Common

TR	7080.0 ms
TE	75.00 ms
TD	0.00 ms
MTC	Off
Magn. Preparation	None
Fat-Water Contrast	Fat Saturation
Fat Saturation	Strong
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Multiple Series	Off
Delay in TR	0.00 ms

Resolution - Common

FOV Read	235 mm

Resolution - Common

FOV Phase	100.0 %
Slice Thickness	2.5 mm
Base Resolution	86
Phase Resolution	72 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	GRAPPA
Reference Scans	GRE/Separate
Acceleration Factor PE	2
Reference Lines PE	24
Phase Partial Fourier	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Hamming	Off
Distortion Correction	Off
Static Field Correction	Off
Normalize	Off

Geometry - Common

Slice Group	1
Slices	60
Distance Factor	0 %
Position	R2.8 A11.0 H5.9 mm
Orientation	Transversal
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FOV Read	235 mm
FOV Phase	100.0 %
Slice Thickness	2.5 mm
TR	7080.0 ms
Multi-Slice Mode	Interleaved
Series	Interleaved
Concatenations	3

Geometry - AutoAlign

Slice Group	1
Position	R2.8 A11.0 H5.9 mm
Orientation	Transversal
Phase Encoding Dir.	A >> P
AutoAlign	---
Initial Position	R2.8 A11.0 H5.9
R	2.8 mm
A	11.0 mm
H	5.9 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

Geometry - Saturation

Special Saturation	None

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Sum of Squares
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Standard
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	R2.8 A11.0 H5.9 mm
Orientation	Transversal
Rotation	0.00 deg
A >> P	235 mm
R >> L	235 mm
F >> H	150 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Standard

System - Tx/Rx

Frequency 1H	123.249408 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	7080.0 ms
Concatenations	3

Inline - Subtraction

Subtract	Off

Inline - Subtraction

Measurements	1
StdDev	Off
Save Original Images	On

Inline - MIP

MIP Sag	Off
MIP Cor	Off
MIP Tra	Off
MIP Time	Off
Radial MIP	Off
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Composing

Inline Composing	Off

Sequence - Part 1

Sequence Name	epse
Excitation	Standard
RF Pulse Type	Normal
Gradient Mode	Fast*
Bandwidth	2326 Hz/Px
Echo Spacing	0.51 ms
Free Echo Spacing	On
EPI Factor	62

Sequence - Part 2

Introduction	Off

Sequence - Assistant

SAR Assistant	Off

\\Research\Investigators\Frederick\CORPUS_CONSISTENT\ep2d_se_sms_mgh
TA: 1 sec Coil Selection: Auto Voxel Size: 3.9×3.9×5.0 mm³ Acc:: None Rel. SNR: 1.00

Properties

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slice Group	1
Slices	1
Distance Factor	50 %
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FOV Read	500 mm
FOV Phase	100.0 %
Slice Thickness	5.0 mm
TR	285.0 ms
TE	182.00 ms
Averages	1
Concatenations	1
AutoAlign	---

Contrast - Common

TR	285.0 ms
TE	182.00 ms
MTC	Off
Magn. Preparation	None
Flip Angle 1	90 deg
Flip Angle 2	180 deg
Fat-Water Contrast	Fat Saturation
Fat Saturation	Weak
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Multiple Series	Off
Delay in TR	0.00 ms

Resolution - Common

FOV Read	500 mm
FOV Phase	100.0 %
Slice Thickness	5.0 mm
Base Resolution	128
Phase Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	None
Phase Partial Fourier	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Hamming	Off
Distortion Correction	2D
Static Field Correction	Off
Normalize	Off

Geometry - Common

Slice Group	1
Slices	1
Distance Factor	50 %
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FOV Read	500 mm
FOV Phase	100.0 %
Slice Thickness	5.0 mm
TR	285.0 ms
Multi-Slice Mode	Interleaved
Series	Interleaved
Concatenations	1

Geometry - AutoAlign

Slice Group	1
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
AutoAlign	---
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

Geometry - Saturation

Special Saturation	None

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Sum of Squares
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Standard
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
A >> P	500 mm
R >> L	500 mm
F >> H	5 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Standard

System - Tx/Rx

Frequency 1H	123.249408 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	285.0 ms
Concatenations	1

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - MIP

MIP Sag	Off
MIP Cor	Off
MIP Tra	Off
MIP Time	Off
Radial MIP	Off
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Composing

Inline Composing	Off

Sequence - Part 1

Sequence Name	epse
Excitation	Standard
RF Pulse Type	Normal
Gradient Mode	Fast
Bandwidth	752 Hz/Px
Echo Spacing	1.38 ms
Free Echo Spacing	Off
EPI Factor	128

Sequence - Part 2

Introduction	Off

Sequence - Special

Imaging Dummy TRs	-1
SMS ACS Dummy TRs	-1
FLEET Dummy Pulses	0
RF Clip	0
VERSE Factor	1.00
ACS mode	Standard
Kernel Size	3x3
Wait for TCP/IP Trigger	Off
Reverse Phase Encoding	Off

Sequence - Assistant

SAR Assistant	Off

\\Research\Investigators\Frederick\CORPUS_CONSISTENT\space_mgh_epinav_ABCD
TA: 1:11 min Coil Selection: Auto Voxel Size: 2.3×2.3×3.0 mm³ Acc:: None Rel. SNR: 1.00

Properties

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slab Group	1
Slabs	1
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slices per Slab	64
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	300 mm
FOV Phase	100.0 %
Slice Thickness	3.00 mm
TR	1000.0 ms
TE	123.00 ms
Averages	1.0
Concatenations	1
AutoAlign	---

Contrast - Common

TR	1000.0 ms
TE	123.00 ms
MTC	Off
Magn. Preparation	None
Flip Angle Mode	T2 Var
Fat-Water Contrast	Standard
Dark Blood	Off
Blood Suppression	Off
Wrap-up Magn.	None
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Multiple Series	Each Measurement
Reordering	Linear

Resolution - Common

FOV Read	300 mm
FOV Phase	100.0 %
Slice Thickness	3.00 mm
Base Resolution	128
Phase Resolution	100 %
Slice Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	None
Phase Partial Fourier	Allowed
Slice Partial Fourier	Off
Elliptical Scanning	On

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	2D
Normalize	Off
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	1
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slices per Slab	64
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	300 mm
FOV Phase	100.0 %
Slice Thickness	3.00 mm
TR	1000.0 ms
Concatenations	1

Geometry - AutoAlign

Slab Group	1
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
AutoAlign	---
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Sagittal
Initial Rotation	0.00 deg

Geometry - Navigator

Geometry - Saturation

Special Saturation	None

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Tune up
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
A >> P	263 mm
R >> L	350 mm
F >> H	350 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Non-sel.

System - Tx/Rx

Frequency 1H	123.249408 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000
Gain	High

Physio - Signal

1st Signal/Mode	None
Trigger Delay	0 ms
TR	1000.0 ms
Concatenations	1

Physio - Cardiac

Fat-Water Contrast	Standard
Magn. Preparation	None
Dark Blood	Off
FOV Read	300 mm
FOV Phase	100.0 %
Phase Resolution	100 %

Physio - PACE

Resp. Control	Off
Concatenations	1

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - MIP

MIP Sag	Off
MIP Cor	Off
MIP Tra	Off
MIP Time	Off
Radial MIP	Off
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Composing

Inline Composing	Off

Sequence - Part 1

Sequence Name	spc
Dimension	3D
Excitation	Non-sel.
RF Pulse Type	Normal
Gradient Mode	Fast
Flow Compensation	None
Reordering	Linear
Bandwidth	501 Hz/Px
Echo Spacing	3.50 ms
Turbo Factor	96
Echo Train Duration	287 ms

Sequence - Part 2

Introduction	On

Sequence - Special

Include Nav.	Off
Apply moco to	parent and nav
Remeasure	0 TRs
Reacq. threshold	0.50
Feedback Delay	80 ms
Moco ref. image	Use Temp Ref

Sequence - Special

K-space streaming	None
ABCD navigator	Off
Apply freq to	parent and nav

Sequence - Assistant

SAR Assistant	Off
Allowed Delay	0 s

\\Research\Investigators\Frederick\CORPUS_CONSISTENT\svs_se
TA: 3:24 min Coil Selection: Auto VoI: 32×20×34 mm³ Rel. SNR: 1.00

Properties

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Position	Isocenter
Orientation	C > T-24.6 > S4.6
Rotation	1.26 deg
VoI R >> L	32 mm
VoI F >> H	20 mm
VoI A >> P	34 mm
TR	3000.0 ms
TE	30.00 ms
Averages	64

Contrast - Common

TR	3000.0 ms
TE	30.00 ms
Flip Angle	90 deg
Preparation Scans	4
Averages	64
Water Suppression	Water Saturation
Water Suppr. BW	35 Hz
Spectral Suppr.	None

Resolution - Common

Vector Size	2048
Normalize	Off

Geometry - Common

Position	Isocenter
Orientation	C > T-24.6 > S4.6
Rotation	1.26 deg
VoI R >> L	32 mm
VoI F >> H	20 mm
VoI A >> P	34 mm

Geometry - AutoAlign

AutoAlign	---

Geometry - AutoAlign

Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

Geometry - Navigator

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Tune up
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off
Adj. Water Suppr.	On

System - Adjust Volume

Position	Isocenter
Orientation	C > T-24.6 > S4.6
Rotation	91.26 deg
F >> H	20 mm
R >> L	32 mm
A >> P	34 mm
Reset	Off

System - pTx

B1 Shim	TrueForm

System - Tx/Rx

Frequency 1H	123.249408 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	3000.0 ms

Physio - PACE

Resp. Control	Off

Sequence - Common

Sequence Name	svs_se
Preparation Scans	4
Delta Frequency	0.00 ppm
Ref. Scan Mode	Off
Measurements	1
Phase Cycling	Auto
Save Uncombined	Off
Bandwidth	1000 Hz
Acquisition Duration	2048 ms
Remove Oversampling	On
Freq. Corr. Accumulation	Off
Incr. Gradient Spoiling	Off

\\Research\Investigators\Frederick\CORPUS_CONSISTENT\svs_st
TA: 6:02 min Coil Selection: Auto VoI: 20×20×20 mm³ Rel. SNR: 1.00

Properties

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
VoI A >> P	20 mm
VoI R >> L	20 mm
VoI F >> H	20 mm
TR	2000.0 ms
TE	20.00 ms
Averages	176

Contrast - Common

TR	2000.0 ms
TE	20.00 ms
TM	10.00 ms
Flip Angle	90 deg
Preparation Scans	4
Averages	176
Water Suppression	Water Saturation
Water Suppr. BW	50 Hz

Resolution - Common

Vector Size	1024
Normalize	Prescan

Geometry - Common

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
VoI A >> P	20 mm
VoI R >> L	20 mm
VoI F >> H	20 mm

Geometry - AutoAlign

AutoAlign	---

Geometry - AutoAlign

Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	When Changed
Assume Silicone	Off
Adj. Water Suppr.	On

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
A >> P	20 mm
R >> L	20 mm
F >> H	20 mm
Reset	Off

System - pTx

B1 Shim	TrueForm

System - Tx/Rx

Frequency 1H	123.249408 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Sequence - Common

Sequence Name	svs_st
Preparation Scans	4
Delta Frequency	-2.30 ppm
Ref. Scan Mode	Inline Correction
No. of Ref. scans	1
Measurements	1

Sequence - Common

Phase Cycling	Auto
Save Uncombined	Off
Bandwidth	1200 Hz
Acquisition Duration	853 ms
Remove Oversampling	On
Freq. Corr. Accumulation	Off

\\Research\Investigators\Frederick\CORPUS_CONSISTENT\tfl
TA: 26 sec Coil Selection: Auto Voxel Size: 2.0×2.0×2.0 mm³ Acc:: None Rel. SNR: 1.00

Properties

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slab Group	1
Slabs	1
Distance Factor	50 %
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slices per Slab	88
Phase Oversampling	0 %
Slice Oversampling	18.2 %
FOV Read	256 mm
FOV Phase	100.0 %
Slice Thickness	2.00 mm
TR	250.0 ms
TE	1.03 ms
Averages	1
Concatenations	1
AutoAlign	Head > Brain

Contrast - Common

TR	250.0 ms
TE	1.03 ms
Magn. Preparation	None
Flip Angle	3 deg
Fat-Water Contrast	Standard
Dark Blood	Off
Contrasts	1
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Multiple Series	Each Measurement
Reordering	Linear

Resolution - Common

FOV Read	256 mm
FOV Phase	100.0 %
Slice Thickness	2.00 mm
Base Resolution	128
Phase Resolution	100 %
Slice Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	None
Phase Partial Fourier	6/8
Slice Partial Fourier	6/8
Asymmetric Echo	Allowed
Elliptical Scanning	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	2D
Normalize	Off
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	1
Distance Factor	50 %
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slices per Slab	88
Phase Oversampling	0 %
Slice Oversampling	18.2 %
FOV Read	256 mm
FOV Phase	100.0 %
Slice Thickness	2.00 mm
TR	250.0 ms
Multi-Slice Mode	Single Shot
Series	Interleaved
Concatenations	1

Geometry - AutoAlign

Slab Group	1
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
AutoAlign	Head > Brain
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm

Geometry - AutoAlign

Initial Orientation	Transversal
Initial Rotation	0.00 deg

Geometry - Navigator

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	ACS All but spine
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Sum of Squares
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Tune up
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
A >> P	263 mm
R >> L	350 mm
F >> H	350 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Non-sel.

System - Tx/Rx

Frequency 1H	123.249408 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	250.0 ms
Concatenations	1

Physio - Cardiac

Fat-Water Contrast	Standard
Magn. Preparation	None
Dark Blood	Off
FOV Read	256 mm
FOV Phase	100.0 %
Phase Resolution	100 %

Physio - PACE

Resp. Control	Off
Concatenations	1

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - MIP

MIP Sag	Off
MIP Cor	Off
MIP Tra	Off
MIP Time	Off
Radial MIP	Off
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Composing

Inline Composing	Off

Sequence - Part 1

Sequence Name	tfl
Dimension	3D
Excitation	Non-sel.
RF Pulse Type	Fast
Gradient Mode	Fast
Flow Compensation	None
Reordering	Linear
Bandwidth	540 Hz/Px
Echo Spacing	3.10 ms
Asymmetric Echo	Allowed
Turbo Factor	78

Sequence - Part 2

Introduction	On
RF Spoiling	On
Incr. Gradient Spoiling	Off

Sequence - Assistant

SAR Assistant	Off

\\Research\Investigators\Frederick\CORPUS_CONSISTENT\tfl_b1map
TA: 11 sec Coil Selection: Auto Voxel Size: 3.1×3.1×8.0 mm³ Acc:: None Rel. SNR: 1.00

Properties

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	On
Load Images to Graphic Segments	On
Graphic segment	Default
Inline Movie	Off

Routine

Slice Group	1
Slices	20
Distance Factor	50 %
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FOV Read	200 mm
FOV Phase	100.0 %
Slice Thickness	8.0 mm
TR	5310.0 ms
TE	1.83 ms
Averages	1
Concatenations	1
AutoAlign	---

Contrast - Common

TR	5310.0 ms
TE	1.83 ms
Magn. Preparation	None
Flip Angle	8 deg
Fat-Water Contrast	Standard
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Multiple Series	Each Measurement

Resolution - Common

FOV Read	200 mm
FOV Phase	100.0 %
Slice Thickness	8.0 mm
Base Resolution	64

Resolution - Common

Phase Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	None
Phase Partial Fourier	Off
Asymmetric Echo	Allowed

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	2D
Normalize	Off
Image Filter	Off

Geometry - Common

Slice Group	1
Slices	20
Distance Factor	50 %
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FOV Read	200 mm
FOV Phase	100.0 %
Slice Thickness	8.0 mm
TR	5310.0 ms
Multi-Slice Mode	Interleaved
Series	Interleaved
Concatenations	1

Geometry - AutoAlign

Slice Group	1
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
AutoAlign	---
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	ACS Restricted
Radial Sorting	Off
MSMA	S - C - T
Sagittal	Med >> Lat
Coronal	A >> P
Transversal	H >> F
Coil Combination	Sum of Squares
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Tune up
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
A >> P	263 mm
R >> L	350 mm
F >> H	350 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Slice-sel.

System - Tx/Rx

Frequency 1H	123.249408 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Sequence - Part 1

Sequence Name	tfl
Dimension	2D
Excitation	Slice-sel.
RF Pulse Type	Low SAR
Gradient Mode	Normal
Flow Compensation	None
Bandwidth	490 Hz/Px
Echo Spacing	4.12 ms
Asymmetric Echo	Allowed
Turbo Factor	64

Sequence - Part 2

Introduction	Off
RF Spoiling	On

Sequence - Assistant

SAR Assistant	Off

\\Research\Investigators\Frederick\CORPUS_CONSISTENT\tfl_mgh_epinav_ABCD
TA: 1:14 min Coil Selection: Auto Voxel Size: 4.0×4.0×4.0 mm³ Acc:: None Rel. SNR: 1.00

Properties

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slab Group	1
Slabs	1
Distance Factor	50 %
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slices per Slab	48
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	256 mm
FOV Phase	100.0 %
Slice Thickness	4.00 mm
TR	1000.0 ms
TE	1.18 ms
Averages	1
Concatenations	1
AutoAlign	---

Contrast - Common

TR	1000.0 ms
TE	1.18 ms
Magn. Preparation	Non-sel. IR
TI	700 ms
Flip Angle	7 deg
Fat-Water Contrast	Standard
Dark Blood	Off
Contrasts	1
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Multiple Series	Each Measurement
Reordering	Linear

Resolution - Common

FOV Read	256 mm
FOV Phase	100.0 %
Slice Thickness	4.00 mm
Base Resolution	64
Phase Resolution	100 %
Slice Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	None
Phase Partial Fourier	Off
Slice Partial Fourier	Off
Asymmetric Echo	Off
Elliptical Scanning	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	Off
Normalize	Off
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	1
Distance Factor	50 %
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slices per Slab	48
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	256 mm
FOV Phase	100.0 %
Slice Thickness	4.00 mm
TR	1000.0 ms
Multi-Slice Mode	Single Shot
Series	Interleaved
Concatenations	1

Geometry - AutoAlign

Slab Group	1
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
AutoAlign	---
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm

Geometry - AutoAlign

Initial Orientation	Sagittal
Initial Rotation	0.00 deg

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Standard
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	Isocenter
Orientation	Sagittal
Rotation	0.00 deg
A >> P	256 mm
F >> H	256 mm
R >> L	192 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Non-sel.

System - Tx/Rx

Frequency 1H	123.249408 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	1000.0 ms
Concatenations	1

Physio - Cardiac

Fat-Water Contrast	Standard
Magn. Preparation	Non-sel. IR
TI	700 ms
Dark Blood	Off
FOV Read	256 mm
FOV Phase	100.0 %
Phase Resolution	100 %

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - MIP

MIP Sag	Off
MIP Cor	Off
MIP Tra	Off
MIP Time	Off
Radial MIP	Off
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Composing

Inline Composing	Off

Sequence - Part 1

Sequence Name	tfl_vn
Dimension	3D
Excitation	Non-sel.
RF Pulse Type	Fast
Gradient Mode	Fast
Flow Compensation	None
Reordering	Linear
Bandwidth	650 Hz/Px
Echo Spacing	2.74 ms
Asymmetric Echo	Off
Turbo Factor	48

Sequence - Part 2

Introduction	Off
RF Spoiling	On
Incr. Gradient Spoiling	Off

Sequence - Special

Readout polarity	Positive
Nav. location	None
Apply moco to	neither
Remeasure	0 TRs
Reacq. threshold	0.50
Feedback Delay	80 ms
Moco ref. image	Use Temp Ref

Sequence - Special

K-space streaming	None
ABCD navigator	Off
Add. grad time	0.00 ms
Apply freq to	neither
Averaging	None

Sequence - Assistant

SAR Assistant	Off

\\Research\Investigators\Frederick\CORPUS_CONSISTENT\tgse_asl
TA: 9:51 min Coil Selection: Auto Voxel Size: 1.5×1.5×3.0 mm³ Acc:: 2 Rel. SNR: 1.00

Properties

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slab Group	1
Slabs	1
Distance Factor	50 %
Position	R1.1 A13.1 H6.4 mm
Orientation	T > C7.0 > S0.8
Phase Encoding Dir.	P >> A
Slices per Slab	40
Phase Oversampling	15 %
Slice Oversampling	25.0 %
FOV Read	192 mm
FOV Phase	100.0 %
Slice Thickness	3.0 mm
TR	5600.0 ms
TE	20.00 ms
Averages	1
Concatenations	1
AutoAlign	Head > Brain

Contrast - Common

TR	5600.0 ms
TE	20.00 ms
Flip Angle	180 deg
Fat-Water Contrast	Fat Saturation
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	10
Multiple Series	Off
Delay in TR	0.00 ms
Reordering	Centric

Contrast - ASL

Perfusion Mode	PCASL

Contrast - ASL

Suppression	Gray-White
Labeling Duration	1800 ms
Postlabeling Delay	1800 ms
Delay Array Size	1

Resolution - Common

FOV Read	192 mm
FOV Phase	100.0 %
Slice Thickness	3.0 mm
Base Resolution	64
Phase Resolution	96 %
Interpolation	On

Resolution - Acceleration

Acceleration Mode	GRAPPA
Reference Scans	GRE/Separate
Acceleration Factor PE	2
Reference Lines PE	24
Acceleration Factor 3D	1
Reference Lines 3D	8
Phase Partial Fourier	7/8
Slice Partial Fourier	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Hamming	Off
Distortion Correction	3D
Normalize	Prescan

Geometry - Common

Slab Group	1
Slabs	1
Distance Factor	50 %
Position	R1.1 A13.1 H6.4 mm
Orientation	T > C7.0 > S0.8
Phase Encoding Dir.	P >> A
Slices per Slab	40
Phase Oversampling	15 %
Slice Oversampling	25.0 %
FOV Read	192 mm
FOV Phase	100.0 %
Slice Thickness	3.0 mm
TR	5600.0 ms
Concatenations	1

Geometry - AutoAlign

Slab Group	1
Position	R1.1 A13.1 H6.4 mm
Orientation	T > C7.0 > S0.8

Geometry - AutoAlign

Phase Encoding Dir.	P >> A
AutoAlign	Head > Brain
Initial Position	Isocenter
R	0.0 mm
A	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	180.00 deg

Geometry - Saturation

Special Saturation	Parallel F
Gap	35.00 mm

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Standard
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	R1.1 A13.1 H6.4 mm
Orientation	T > C7.0 > S0.8
Rotation	-175.40 deg
A >> P	192 mm
R >> L	192 mm
F >> H	120 mm
Reset	Off

System - pTx

B1 Shim	TrueForm

System - Tx/Rx

Frequency 1H	123.249408 MHz
? Ref. Amplitude 1H	0.000 V

System - Tx/Rx

Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	5600.0 ms
Segments	5
Concatenations	1

Sequence - Part 1

Sequence Name	tgse
Dimension	3D
RF Pulse Type	Normal
Gradient Mode	Fast
Reordering	Centric
Bandwidth	2232 Hz/Px
Echo Spacing	0.53 ms
Turbo Factor	10
Segments	5
EPI Factor	31

Sequence - Part 2

Introduction	Off

Sequence - Assistant

SAR Assistant	Off

\\Research\Investigators\Frederick\CORPUS_CONSISTENT\tse
TA: 3:32 min Coil Selection: Auto Voxel Size: 0.4×0.4×4.0 mm³ Acc:: None Rel. SNR: 1.00

Properties

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slice Group	1
Slices	30
Distance Factor	25 %
Position	L0.0 P0.0 H0.0 mm
Orientation	Transversal
Phase Encoding Dir.	R >> L
Phase Oversampling	0 %
FOV Read	210 mm
FOV Phase	81.3 %
Slice Thickness	4.0 mm
TR	3000.0 ms
TE 1	9.00 ms
TE 2	81.00 ms
Averages	1
Concatenations	2
AutoAlign	Head > Brain

Contrast - Common

TR	3000.0 ms
TE 1	9.00 ms
TE 2	81.00 ms
TD	0.00 ms
MTC	Off
Magn. Preparation	None
Flip Angle Mode	Constant
Flip Angle	120 deg
Fat-Water Contrast	Standard
Dark Blood	Off
Contrasts	2
Wrap-up Magn.	None
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Multiple Series	Each Measurement

Resolution - Common

FOV Read	210 mm
FOV Phase	81.3 %
Slice Thickness	4.0 mm
Base Resolution	256
Phase Resolution	80 %
Interpolation	On

Resolution - Acceleration

Acceleration Mode	None
Deep Resolve	Off
Phase Partial Fourier	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	2D
Normalize	Prescan
Image Filter	Off

Geometry - Common

Slice Group	1
Slices	30
Distance Factor	25 %
Position	L0.0 P0.0 H0.0 mm
Orientation	Transversal
Phase Encoding Dir.	R >> L
Phase Oversampling	0 %
FOV Read	210 mm
FOV Phase	81.3 %
Slice Thickness	4.0 mm
TR	3000.0 ms
Multi-Slice Mode	Interleaved
Series	Interleaved
Concatenations	2

Geometry - AutoAlign

Slice Group	1
Position	L0.0 P0.0 H0.0 mm
Orientation	Transversal
Phase Encoding Dir.	R >> L
AutoAlign	Head > Brain
Initial Position	L0.0 P0.0 F37.0
L	0.0 mm
P	0.0 mm
H	37.0 mm
Initial Orientation	Transversal
Initial Rotation	90.00 deg

Geometry - Navigator

Geometry - Saturation

Special Saturation	None

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	37 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	ACS All but spine
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Standard
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	L0.0 P0.0 H0.0 mm
Orientation	Transversal
Rotation	90.00 deg
R >> L	171 mm
A >> P	210 mm
F >> H	149 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
LR Balancing	Off

System - Tx/Rx

Frequency 1H	123.249408 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	3000.0 ms
Concatenations	2

Physio - Cardiac

Fat-Water Contrast	Standard
Magn. Preparation	None
Dark Blood	Off
FOV Read	210 mm
FOV Phase	81.3 %
Phase Resolution	80 %
Motion Correction	None

Physio - PACE

Resp. Control	Off
Concatenations	2

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Motion Correction	None
Save Original Images	On

Inline - MIP

MIP Sag	Off
MIP Cor	Off
MIP Tra	Off
MIP Time	Off
Radial MIP	Off
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Composing

Inline Composing	Off

Sequence - Part 1

Sequence Name	tse
Dimension	2D
RF Pulse Type	Normal
Gradient Mode	Normal
Flow Compensation	None
Bandwidth	250 Hz/Px
Echo Spacing	8.96 ms
Free Echo Spacing	Off
Define	Turbo Factor
Turbo Factor	5
Echo Trains per Slice	34

Sequence - Part 2

Introduction	On
Phase Correction	Automatic
Compensate T2 Decay	Off
Fast Mode	Off
WARP	Off
Red. EC Sensitivity	Off
Acoustic noise reduction	Off

Sequence - Part 2

Reduce Motion Sens.	On
Motion Correction	None

Sequence - Assistant

SAR Assistant	Flip Angle
Min Flip Angle	130 deg
Allowed Delay	60 s

\\Research\Investigators\Frederick\CORPUS_CONSISTENT\tse_MDME
TA: 4:25 min Coil Selection: Auto Voxel Size: 0.7×0.7×4.0 mm³ Acc:: 3 Rel. SNR: 1.00

Properties

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	On
Load Images to Graphic Segments	On
Graphic segment	Default
Inline Movie	Off

Routine

Slice Group	1
Slices	30
Distance Factor	25 %
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	R >> L
Phase Oversampling	0 %
FOV Read	230 mm
FOV Phase	81.3 %
Slice Thickness	4.0 mm
TR	4350.0 ms
TE 1	23.00 ms
TE 2	101.00 ms
Averages	1
AutoAlign	---

Contrast - Common

TR	4350.0 ms
TE 1	23.00 ms
TE 2	101.00 ms
Magn. Preparation	Slice-sel. IR
Flip Angle	150 deg
Fat-Water Contrast	Standard
Contrasts	2
Wrap-up Magn.	None
Reconstruction	Magn./Phase

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	4
Pause after Meas. 1	0.0 s
Pause after Meas. 2	0.0 s
Pause after Meas. 3	0.0 s
Multiple Series	Off

Resolution - Common

FOV Read	230 mm
FOV Phase	81.3 %
Slice Thickness	4.0 mm
Base Resolution	320
Phase Resolution	80 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	GRAPPA
Reference Scans	TSE/Separate
Acceleration Factor PE	3
Reference Lines PE	32

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	2D
Normalize	Prescan
Image Filter	Off

Geometry - Common

Slice Group	1
Slices	30
Distance Factor	25 %
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	R >> L
Phase Oversampling	0 %
FOV Read	230 mm
FOV Phase	81.3 %
Slice Thickness	4.0 mm
TR	4350.0 ms
Multi-Slice Mode	Interleaved

Geometry - AutoAlign

Slice Group	1
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	R >> L
AutoAlign	---
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	90.00 deg

Geometry - Saturation

Special Saturation	None

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Tune up
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
A >> P	263 mm
R >> L	350 mm
F >> H	350 mm
Reset	Off

System - pTx

B1 Shim	TrueForm

System - Tx/Rx

Frequency 1H	123.249408 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Inline - Subtraction

Subtract	Off
Measurements	4
StdDev	Off
Save Original Images	On

Inline - MIP

MIP Sag	Off
MIP Cor	Off
MIP Tra	Off
MIP Time	Off

Inline - MIP

Radial MIP	Off
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Composing

Inline Composing	Off

Sequence - Part 1

Sequence Name	MDME
Dimension	2D
RF Pulse Type	Normal
Gradient Mode	Fast
Flow Compensation	None
Bandwidth	150 Hz/Px
Echo Spacing	11.26 ms
Define	Turbo Factor
Turbo Factor	5

Sequence - Part 2

Phase Correction	Automatic
Reduce Motion Sens.	Off

Sequence - Assistant

SAR Assistant	Off

\\Research\Investigators\Frederick\CORPUS_CONSISTENT\tse_dixon
TA: 5:03 min Coil Selection: Auto Voxel Size: 0.5×0.5×3.0 mm³ Acc:: 2 Rel. SNR: 1.00

Properties

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	On
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slice Group	1
Slices	40
Distance Factor	33 %
Position	R105.1 A10.0 H6.0 mm
Orientation	S > T-1.6
Phase Encoding Dir.	H >> F
Phase Oversampling	30 %
FOV Read	250 mm
FOV Phase	200.0 %
Slice Thickness	3.0 mm
TR	4300.0 ms
TE	81.00 ms
Averages	1
Concatenations	2
AutoAlign	---

Contrast - Common

TR	4300.0 ms
TE	81.00 ms
TD	0.00 ms
MTC	Off
Magn. Preparation	None
Flip Angle Mode	Constant
Flip Angle	150 deg
Fat-Water Contrast	Dixon
Fat Saturation	Weak
Dark Blood	Off
Contrasts	1
Wrap-up Magn.	Restore
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Multiple Series	Each Measurement

Resolution - Common

FOV Read	250 mm
FOV Phase	200.0 %
Slice Thickness	3.0 mm
Base Resolution	256
Phase Resolution	80 %
Trajectory	Cartesian
Interpolation	On

Resolution - Acceleration

Acceleration Mode	GRAPPA
Reference Scans	Integrated
Acceleration Factor PE	2
Reference Lines PE	43
Deep Resolve	Off
Phase Partial Fourier	Off
Asymmetric Echo	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	2D
Normalize	Prescan
Image Filter	On

Geometry - Common

Slice Group	1
Slices	40
Distance Factor	33 %
Position	R105.1 A10.0 H6.0 mm
Orientation	S > T-1.6
Phase Encoding Dir.	H >> F
Phase Oversampling	30 %
FOV Read	250 mm
FOV Phase	200.0 %
Slice Thickness	3.0 mm
TR	4300.0 ms
Multi-Slice Mode	Interleaved
Series	Interleaved
Concatenations	2

Geometry - AutoAlign

Slice Group	1
Position	R105.1 A10.0 H6.0 mm
Orientation	S > T-1.6
Phase Encoding Dir.	H >> F
AutoAlign	---
Initial Position	R105.1 A10.0 H6.0
R	105.1 mm
A	10.0 mm
H	6.0 mm

Geometry - AutoAlign

Initial Orientation	S > T
S > T	-1.60
> C	0.00
Initial Rotation	90.00 deg

Geometry - Navigator

Geometry - Saturation

Special Saturation	None

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	6 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	ACS All but spine
Radial Sorting	Off
MSMA	S - C - T
Sagittal	L >> R
Coronal	P >> A
Transversal	H >> F
Coil Combination	Adaptive Combine
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Standard
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	R105.1 A10.0 H6.0 mm
Orientation	S > T-1.6
Rotation	180.00 deg
A >> P	250 mm
F >> H	500 mm
R >> L	159 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
LR Balancing	Off

System - Tx/Rx

Frequency 1H	123.249408 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	4300.0 ms
Concatenations	2

Physio - Cardiac

Fat-Water Contrast	Dixon
Magn. Preparation	None
Dark Blood	Off
FOV Read	250 mm
FOV Phase	200.0 %
Phase Resolution	80 %
Trajectory	Cartesian

Physio - PACE

Resp. Control	Off
Concatenations	2

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - MIP

MIP Sag	Off
MIP Cor	Off
MIP Tra	Off
MIP Time	Off
Radial MIP	Off
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Composing

Inline Composing	Off

Sequence - Part 1

Sequence Name	tseR
RF Pulse Type	Normal
Gradient Mode	Normal
Flow Compensation	None
Bandwidth	349 Hz/Px
Echo Spacing	11.64 ms
Asymmetric Echo	Off
Define	Turbo Factor
Turbo Factor	17
Echo Trains per Slice	17

Sequence - Part 2

Introduction	On
Phase Correction	Automatic
Red. EC Sensitivity	Off
Reduce Motion Sens.	On

Sequence - Assistant

SAR Assistant	Flip Angle
Min Flip Angle	130 deg
Allowed Delay	30 s

\\Research\Investigators\Frederick\CORPUS_CONSISTENT\twist
TA: 1:32 min Coil Selection: Auto Voxel Size: 1.0×1.0×1.0 mm³ Acc:: 6 Rel. SNR: 1.00

Properties

Start measurement without further preparation	On
Wait for User to Start	On
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	On
Auto Close Inline Display	On
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slab Group	1
Slabs	1
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slices per Slab	160
Phase Oversampling	0 %
Slice Oversampling	20.0 %
FOV Read	256 mm
FOV Phase	100.0 %
Slice Thickness	1.00 mm
TR	2.26 ms
TE	0.92 ms
AutoAlign	---

Contrast - Common

TR	2.26 ms
TE	0.92 ms
Flip Angle	25 deg
Fat-Water Contrast	Standard
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	TWIST
Central Region A	15 %
Sampling Density B	20 %
Reconstruction Scheme	Symmetric Share
Temporal Resolution	1.99 s
Temporal Interpolation	1
Measurements	40
Pause after Meas. 1	0.0 s
Pause after Meas. 2	0.0 s
Pause after Meas. 3	0.0 s
Pause after Meas. 4	0.0 s
Pause after Meas. 5	0.0 s

Contrast - Dynamic

Pause after Meas. 6	0.0 s
Pause after Meas. 7	0.0 s
Pause after Meas. 8	0.0 s
Pause after Meas. 9	0.0 s
Pause after Meas. 10	0.0 s
Pause after Meas. 11	0.0 s
Pause after Meas. 12	0.0 s
Pause after Meas. 13	0.0 s
Pause after Meas. 14	0.0 s
Pause after Meas. 15	0.0 s
Pause after Meas. 16	0.0 s
Pause after Meas. 17	0.0 s
Pause after Meas. 18	0.0 s
Pause after Meas. 19	0.0 s
Pause after Meas. 20	0.0 s
Pause after Meas. 21	0.0 s
Pause after Meas. 22	0.0 s
Pause after Meas. 23	0.0 s
Pause after Meas. 24	0.0 s
Pause after Meas. 25	0.0 s
Pause after Meas. 26	0.0 s
Pause after Meas. 27	0.0 s
Pause after Meas. 28	0.0 s
Pause after Meas. 29	0.0 s
Pause after Meas. 30	0.0 s
Pause after Meas. 31	0.0 s
Pause after Meas. 32	0.0 s
Pause after Meas. 33	0.0 s
Pause after Meas. 34	0.0 s
Pause after Meas. 35	0.0 s
Pause after Meas. 36	0.0 s
Pause after Meas. 37	0.0 s
Pause after Meas. 38	0.0 s
Pause after Meas. 39	0.0 s
Multiple Series	Off
Time to Center	8.5 s
Burn Time to Center	Off

Resolution - Common

FOV Read	256 mm
FOV Phase	100.0 %
Slice Thickness	1.00 mm
Base Resolution	256
Phase Resolution	70 %
Slice Resolution	75 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	GRAPPA
Reference Scans	GRE/Separate

Resolution - Acceleration

Acceleration Factor PE	2
Reference Lines PE	24
Acceleration Factor 3D	3
Reference Lines 3D	24
Phase Partial Fourier	6/8
Slice Partial Fourier	6/8
Asymmetric Echo	Strong
Elliptical Scanning	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	On
Distortion Correction	2D
Normalize	Prescan
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	1
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slices per Slab	160
Phase Oversampling	0 %
Slice Oversampling	20.0 %
FOV Read	256 mm
FOV Phase	100.0 %
Slice Thickness	1.00 mm
TR	2.26 ms

Geometry - AutoAlign

Slab Group	1
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
AutoAlign	---
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Sagittal
Initial Rotation	0.00 deg

Geometry - Saturation

Special Saturation	None

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	Auto Coil Select

System - Miscellaneous

Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Tune up
B1 Shim	TrueForm
Adjustment Tolerance	Maximum
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
A >> P	263 mm
R >> L	350 mm
F >> H	350 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Non-sel.

System - Tx/Rx

Frequency 1H	123.249408 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Inline - Subtraction

Subtract	On
Subtraction Mode	Standard
Save Subtracted Images	On
Subtrahend	1
Subtraction Group	1
Measurements	40
Subtraction Indices	
Autoscaling	Off
Scaling Factor	3
Offset	0
StdDev	Off
Save Original Images	On

Inline - MIP

MIP Sag	Off
MIP Cor	On

Inline - MIP

MIP Tra	Off
MIP Time	Off
Radial MIP	Off
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Composing

Inline Composing	Off

Sequence - Part 1

Sequence Name	fldyn
Dimension	3D
Excitation	Non-sel.
RF Pulse Type	Normal
Gradient Mode	Normal
Bandwidth	750 Hz/Px
Asymmetric Echo	Strong
Optimization	Min. TE TR

Sequence - Part 2

RF Spoiling	On
Phase Enc. Rewinder	On

Sequence - Assistant

SAR Assistant	Flip Angle
Min Flip Angle	15 deg
Allowed Delay	0 s
Optimization	Min. TE TR

\\Research\Investigators\Frederick\CORPUS_CONSISTENT\AALScout
TA: 14 sec Coil Selection: Auto Voxel Size: 1.6×1.6×1.6 mm³ Acc:: 3 Rel. SNR: 1.00

Properties

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	On
Graphic segment	Default
Inline Movie	Off

Routine

Slab Group	1
Slabs	1
Distance Factor	20 %
Position	L0.0 A10.0 H0.0 mm
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slices per Slab	128
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	260 mm
FOV Phase	100.0 %
Slice Thickness	1.6 mm
TR	3.15 ms
TE	1.37 ms
Averages	1
Concatenations	1
AutoAlign	Head

Contrast - Common

TR	3.15 ms
TE	1.37 ms
Flip Angle	8 deg
Fat-Water Contrast	Standard
Contrasts	1
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Time to Center	6.2 s

Resolution - Common

FOV Read	260 mm
FOV Phase	100.0 %

Resolution - Common

Slice Thickness	1.6 mm
Base Resolution	160
Phase Resolution	100 %
Slice Resolution	69 %
Trajectory	Cartesian

Resolution - Acceleration

Acceleration Mode	GRAPPA
Reference Scans	Integrated
Acceleration Factor PE	3
Reference Lines PE	24
Acceleration Factor 3D	1
Phase Partial Fourier	6/8
Slice Partial Fourier	6/8
Asymmetric Echo	Weak

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	3D
Normalize	Prescan
Noise Masking	Off
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	1
Distance Factor	20 %
Position	L0.0 A10.0 H0.0 mm
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slices per Slab	128
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	260 mm
FOV Phase	100.0 %
Slice Thickness	1.6 mm
TR	3.15 ms
Multi-Slice Mode	Sequential
Series	Ascending
Concatenations	1

Geometry - AutoAlign

Slab Group	1
Position	L0.0 A10.0 H0.0 mm
Orientation	Sagittal
Phase Encoding Dir.	A >> P
AutoAlign	Head
Initial Position	Isocenter
L	0.0 mm

Geometry - AutoAlign

P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	ACS All but spine
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Tune up
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
A >> P	263 mm
R >> L	350 mm
F >> H	350 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Non-sel.

System - Tx/Rx

Frequency 1H	123.249408 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - PACE

Resp. Control	Off
Concatenations	1

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - MIP

MIP Sag	Off
MIP Cor	Off
MIP Tra	Off
MIP Time	Off
Radial MIP	Off
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Composing

Inline Composing	Off

Sequence - Part 1

Sequence Name	fl
Dimension	3D
Excitation	Non-sel.
RF Pulse Type	Fast
Gradient Mode	Normal
Bandwidth	540 Hz/Px
Asymmetric Echo	Weak

Sequence - Part 2

Introduction	On
RF Spoiling	On
Breast Application	Off
Phase Enc. Order	Automatic

Sequence - Assistant

SAR Assistant	Off

\\Research\Investigators\Frederick\CORPUS_CONSISTENT\BEAT
TA: 31:18 min Coil Selection: Auto Voxel Size: 0.5×0.5×0.3 mm³ Acc:: None Rel. SNR: 1.00

Properties

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slab Group	1
Slabs	8
Distance Factor	-22 %
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	R >> L
Slices per Slab	64
Phase Oversampling	0 %
Slice Oversampling	12.5 %
FOV Read	200 mm
FOV Phase	91.2 %
Slice Thickness	0.30 mm
TR	20.31 ms
TE	4.30 ms
Averages	1
Concatenations	8
AutoAlign	Head > Brain

Contrast - Common

TR	20.31 ms
TE	4.30 ms
TD	0.00 ms
MTC	Off
Magn. Preparation	None
Flip Angle Mode	Constant
Flip Angle	20 deg
Fat-Water Contrast	Standard
Dark Blood	Off
Contrasts	1
Wrap-up Magn.	None
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1

Contrast - Dynamic

Multiple Series	Off
Reordering	Linear

Contrast - Angio

Flow Direction	F >> H
TONE Ramp	60 %

Resolution - Common

FOV Read	200 mm
FOV Phase	91.2 %
Slice Thickness	0.30 mm
Base Resolution	272
Phase Resolution	100 %
Slice Resolution	64 %
Trajectory	Cartesian
Interpolation	1.50

Resolution - Acceleration

Acceleration Mode	None
Phase Partial Fourier	Off
Slice Partial Fourier	Off
Asymmetric Echo	Weak
Elliptical Scanning	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
POCS	Off
Distortion Correction	2D
Normalize	Prescan
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	8
Distance Factor	-22 %
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	R >> L
Slices per Slab	64
Phase Oversampling	0 %
Slice Oversampling	12.5 %
FOV Read	200 mm
FOV Phase	91.2 %
Slice Thickness	0.30 mm
TR	20.31 ms
Multi-Slice Mode	Sequential
Series	Descending
Concatenations	8

Geometry - AutoAlign

Slab Group	1
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	R >> L
AutoAlign	Head > Brain
Initial Position	Isocenter
L	0.0 mm
A	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	90.00 deg

Geometry - Navigator

Geometry - Saturation

Special Saturation	Tracking H
Gap	10.00 mm
Thickness	40.00 mm

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Performance
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Tune up
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
A >> P	263 mm
R >> L	350 mm
F >> H	350 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	TONE

System - Tx/Rx

Frequency 1H	123.249408 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	20.31 ms
Segments	1
Concatenations	8

Physio - Cardiac

Fat-Water Contrast	Standard
Magn. Preparation	None
Dark Blood	Off
FOV Read	200 mm
FOV Phase	91.2 %
Phase Resolution	100 %
Cine	Off
Trajectory	Cartesian
Dummy Heartbeats	1

Physio - PACE

Resp. Control	Off
Concatenations	8

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - Cardiac

Inline Evaluation	Off
Magn. Preparation	None
Save Original Images	On
Contrasts	1
TE	4.30 ms
TR	20.31 ms

Inline - MIP

MIP Sag	On
MIP Cor	On
MIP Tra	On
MIP Time	Off
Radial MIP	On
Number of Radial Views	12
Axis of Radial Views	H-F
Save Original Images	On
MPR Sag	Off

Inline - MIP

MPR Cor	Off
MPR Tra	Off

Inline - Composing

Inline Composing	Off

Sequence - Part 1

Sequence Name	fl_r
Dimension	3D
Sequence Type	Gre
Excitation	TONE
RF Pulse Type	Normal
Gradient Mode	Fast
Flow Compensation	Slice/Read
Reordering	Linear
Bandwidth	188 Hz/Px
Echo Spacing	9.66 ms
Asymmetric Echo	Weak
Optimization	None
Define	Segments
Segments	1

Sequence - Part 2

Introduction	Off
RF Spoiling	On
Phase Enc. Rewinder	On

Sequence - Assistant

SAR Assistant	TR
Max. TR	21.0 ms
Allowed Delay	0 s
Optimization	None

\\Research\Investigators\Frederick\CORPUS_CONSISTENT\CMRR_XFL_mbPCASL
TA: 5:29 min Coil Selection: Auto Voxel Size: 2.5×2.5×2.3 mm³ Acc:: 6 Rel. SNR: 1.00

Properties

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slice Group	1
Slices	60
Distance Factor	10 %
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FOV Read	215 mm
FOV Phase	100.0 %
Slice Thickness	2.3 mm
TR	3580.0 ms
TE	19.00 ms
Averages	1
Concatenations	1
AutoAlign	Head > Brain

Contrast - Common

TR	3580.0 ms
TE	19.00 ms
MTC	Off
Flip Angle	90 deg
Fat-Water Contrast	Fat Saturation
Fat Saturation	Strong
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	90
Delay in TR	0.00 ms

Resolution - Common

FOV Read	215 mm
FOV Phase	100.0 %
Slice Thickness	2.3 mm

Resolution - Common

Base Resolution	86
Phase Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	SMS
Reference Scans	EPI/Separate
Acceleration Factor PE	1
SMS Factor	6
Phase Partial Fourier	6/8

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Hamming	Off
Distortion Correction	2D
Static Field Correction	Off
Normalize	Off

Geometry - Common

Slice Group	1
Slices	60
Distance Factor	10 %
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FOV Read	215 mm
FOV Phase	100.0 %
Slice Thickness	2.3 mm
TR	3580.0 ms
Multi-Slice Mode	Interleaved
Series	Ascending
Concatenations	1

Geometry - AutoAlign

Slice Group	1
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
AutoAlign	Head > Brain
Initial Position	L0.0 P0.0 F4.0
L	0.0 mm
P	0.0 mm
F	4.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

Geometry - Saturation

Special Saturation	None

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	4 mm
Table Position	F
Inline Composing	Off

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Sum of Squares
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Advanced
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
A >> P	215 mm
R >> L	215 mm
F >> H	150 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Standard

System - Tx/Rx

Frequency 1H	123.249408 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	3580.0 ms
Log Signals	Off
Concatenations	1

BOLD

GLM Statistics	Off
Ignore Meas. at Start	0
Ignore After Transition	0

BOLD

Model Transition States	Off
Temp. Highpass Filter	Off
Threshold	4.00
Paradigm Size	3
Meas[1]	Baseline
Meas[2]	Baseline
Meas[3]	Active
Motion Correction	Off
Spatial Filter	Off
Measurements	90
Delay in TR	0.00 ms

Sequence - Part 1

Sequence Name	epfid
Excitation	Standard
RF Pulse Type	Normal
Gradient Mode	Fast
Bandwidth	2326 Hz/Px
Echo Spacing	0.57 ms
Free Echo Spacing	Off
EPI Factor	86

Sequence - Part 2

Introduction	Off

Sequence - Special

PCASL Options	mdPCASL
Labeling Duration.	1500000 us.
Postlabeling delay.	1641860 us.

Sequence - Assistant

SAR Assistant	Off

\\Research\Investigators\Frederick\CORPUS_CONSISTENT\NoiseSensitivityMap
TA: 5 sec Coil Selection: Auto Voxel Size: 1.2×1.2×5.0 mm³ Rel. SNR: 1.00

Properties

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slice Group	1
Slices	1
Distance Factor	20 %
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FOV Read	300 mm
FOV Phase	100.0 %
Slice Thickness	5.0 mm
TR	9.1 ms
TE	4.80 ms
Averages	1
Concatenations	1
AutoAlign	---

Contrast - Common

TR	9.1 ms
TE	4.80 ms
MTC	Off
Flip Angle	15 deg
Fat-Water Contrast	Standard
Contrasts	1
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Multiple Series	Off

Resolution - Common

FOV Read	300 mm
FOV Phase	100.0 %
Slice Thickness	5.0 mm

Resolution - Common

Base Resolution	256
Phase Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Phase Partial Fourier	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	2D
Normalize	Off
Image Filter	Off

Geometry - Common

Slice Group	1
Slices	1
Distance Factor	20 %
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FOV Read	300 mm
FOV Phase	100.0 %
Slice Thickness	5.0 mm
TR	9.1 ms
Multi-Slice Mode	Sequential
Series	Ascending
Concatenations	1

Geometry - AutoAlign

Slice Group	1
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
AutoAlign	---
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

Geometry - Saturation

Special Saturation	None

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H

Geometry - Tim Planning Suite

Inline Composing	Off

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Tune up
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
A >> P	263 mm
R >> L	350 mm
F >> H	350 mm
Reset	Off

System - pTx

B1 Shim	TrueForm

System - Tx/Rx

Frequency 1H	123.249408 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	9.1 ms
Concatenations	1

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - MIP

MIP Sag	Off
MIP Cor	Off

Inline - MIP

MIP Tra	Off
MIP Time	Off
Radial MIP	Off
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Composing

Inline Composing	Off

Sequence - Part 1

Sequence Name	fl
Dimension	2D
Gradient Mode	Fast
Bandwidth	390 Hz/Px

Sequence - Part 2

Introduction	Off
RF Spoiling	On

Sequence - Nuclei

TX/RX Nucleus	1H
TX/RX Delta Frequency	0 Hz
TX Nucleus	None
TX Delta Frequency	0 Hz

Sequence - Special

ICE program	CoilArrayUtil
number of noise lines	384 lines
Optimal SNR	On
GFactor	On
Condition number	Off
Rx coil diode switching	On
coil channel reordering	Off

Sequence - Assistant

SAR Assistant	Off

\\Research\Investigators\Frederick\CORPUS_CONSISTENT\csi_st
TA: 11:20 min Coil Selection: Auto Voxel Size: 10.0×10.0×15.0 mm³ Rel. SNR: 1.00

Properties

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
FOV A >> P	160 mm
FOV R >> L	160 mm
Thickness F >> H	15 mm
Vol A >> P	80 mm
Vol R >> L	80 mm
TR	1700.0 ms
TE	20.00 ms
Averages	5

Contrast - Common

TR	1700.0 ms
TE	20.00 ms
TM	10.00 ms
Flip Angle	90 deg
Preparation Scans	4
Averages	5
Water Suppression	Water Saturation
Water Suppr. BW	50 Hz

Resolution - Common

FOV A >> P	160 mm
FOV R >> L	160 mm
Thickness F >> H	15 mm
Scan Res. A >> P	16
Scan Res. R >> L	16
Interpol. Res. A >> P	16
Interpol. Res. R >> L	16
Hamming	On
Dimension	2D
Phase Encoding	Weighted
Vector Size	1024
Normalize	Prescan

Geometry - Common

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
FOV A >> P	160 mm
FOV R >> L	160 mm
Thickness F >> H	15 mm
Vol A >> P	80 mm
Vol R >> L	80 mm

Geometry - AutoAlign

Slice Group	1
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
AutoAlign	---
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	When Changed
Assume Silicone	Off
Adj. Water Suppr.	On

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
A >> P	80 mm
R >> L	80 mm
F >> H	15 mm
Reset	Off

System - pTx

B1 Shim	TrueForm

System - Tx/Rx

Frequency 1H	123.249408 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Sequence - Common

Sequence Name	csi_st
Preparation Scans	4
Delta Frequency	-2.70 ppm
Measurements	1
Dimension	2D
Save Uncombined	Off
Bandwidth	1200 Hz
Acquisition Duration	853 ms
Remove Oversampling	On

\\Research\Investigators\Frederick\CORPUS_CONSISTENT\dkd_svs_sLASER
TA: 15 sec Coil Selection: Auto Vol: 20×20×20 mm^3 Rel. SNR: 1.00

Properties

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
Vol A >> P	20 mm
Vol R >> L	20 mm
Vol F >> H	20 mm
TR	5000.0 ms
TE 1	8.00 ms
TE 2	12.00 ms
TE 3	10.00 ms
Averages	1

Contrast - Common

TR	5000.0 ms
TE 1	8.00 ms
TE 2	12.00 ms
TE 3	10.00 ms
Total TE	30.00 ms
Excitation Flip angle	90 deg
Refocusing Flip angle	180 deg
Preparation Scans	2
Averages	1
VAPOR WS	Off

Resolution - Common

Vector Size	2048
Normalize	Off

Geometry - Common

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
Vol A >> P	20 mm
Vol R >> L	20 mm

Geometry - Common

Vol F >> H	20 mm

Geometry - AutoAlign

AutoAlign	---
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

Geometry - Navigator

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Advanced
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
A >> P	20 mm
R >> L	20 mm
F >> H	20 mm
Reset	Off

System - pTx

B1 Shim	TrueForm

System - Tx/Rx

Frequency 1H	123.249408 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None

Physio - Signal

TR	5000.0 ms

Sequence - Common

Sequence Name	slaser
Preparation Scans	2
Delta Frequency	0.00 ppm
Phase Cycling	Auto
Save Uncombined	Off
Bandwidth	6000 Hz
Acquisition Duration	341 ms
Remove Oversampling	On

Sequence - Special

Excitation duration	2000.00 us
Refocusing duration	4500.00 us
AFP Type	GOIA-WURST
Bandwidth_1ms	45 kHz
HSn modulation	16
Gradient factor	85 %
AutoShim	Off
GOIA/FOCI	On
Inversion pulse	Off
FA AutoCalib	Off
AutoVOI	Off
Advanced User	Off
Calibration Type	None
Debug Type	None
Spinal Cord Navigator	Off
Gradient Max. Amplitude	37 mT/m
Ramp time	200 us

\\Research\Investigators\Frederick\CORPUS_CONSISTENT\eja_csi_fid
TA: 1:42:33 h Coil Selection: Auto Voxel Size: 12.5×12.5×25.0 mm³ Rel. SNR: 1.00

Properties

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
FOV A >> P	200 mm
FOV R >> L	200 mm
FOV F >> H	200 mm
TR	3000.0 ms
TE	5.00 ms
Averages	1

Contrast - Common

TR	3000.0 ms
TE	5.00 ms
Flip Angle	90 deg
Preparation Scans	0
Averages	1
Water Suppression	None

Resolution - Common

FOV A >> P	200 mm
FOV R >> L	200 mm
FOV F >> H	200 mm
Scan Res. A >> P	16
Scan Res. R >> L	16
Scan Res. F >> H	8
Interpol. Res. A >> P	16
Interpol. Res. R >> L	16
Interpol. Res. F >> H	8
Hamming	Off
Dimension	3D
Phase Encoding	Full
Vector Size	2048
Normalize	Off

Geometry - Common

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
FOV A >> P	200 mm
FOV R >> L	200 mm
FOV F >> H	200 mm

Geometry - AutoAlign

Slab Group	1
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Advanced
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off
Adj. Water Suppr.	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
A >> P	200 mm
R >> L	200 mm
F >> H	200 mm
Reset	Off

System - pTx

B1 Shim	TrueForm

System - Tx/Rx

Frequency 1H	123.249408 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	3000.0 ms

Sequence - Common

Sequence Name	fid
Preparation Scans	0
Delta Frequency	0.00 ppm
Measurements	1
Dimension	3D
Phase Cycling	None
Save Uncombined	Off
Bandwidth	2000 Hz
Acquisition Duration	1024 ms
Remove Oversampling	On

Sequence - Nuclei

TX/RX Nucleus	1H
TX/RX Delta Frequency	0 Hz
TX Nucleus	None
TX Delta Frequency	0 Hz

Sequence - Special

Excite pulse duration	1000 us
Acq. window shift	200 us
Min. settling delay	300 us
Gradient ramp time	140 us
Use expt. SAR calc	Off
Check RF clipping	Off
Send ref. scans	Off
Inversion pulse	Off
Shift RO frequency	Off
Debug loop type	None

\\Research\Investigators\Frederick\CORPUS_CONSISTENT\eja_fid
TA: 12 sec Coil Selection: Auto Rel. SNR: 1.00

Properties

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

TR	3000.0 ms
TE	5.00 ms
Averages	1

Contrast - Common

TR	3000.0 ms
TE	5.00 ms
Flip Angle	90 deg
Preparation Scans	0
Averages	1
Water Suppression	None

Resolution - Common

Vector Size	2048
Normalize	Off

Geometry - AutoAlign

Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

Geometry - Saturation

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Advanced
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off
Adj. Water Suppr.	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
A >> P	500 mm
R >> L	500 mm
F >> H	500 mm
Reset	Off

System - pTx

B1 Shim	TrueForm

System - Tx/Rx

Frequency 1H	123.249408 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	3000.0 ms

Sequence - Common

Sequence Name	fid
Preparation Scans	0
Delta Frequency	0.00 ppm
Measurements	1
Phase Cycling	None
Save Uncombined	Off
Bandwidth	2000 Hz
Acquisition Duration	1024 ms
Remove Oversampling	On

Sequence - Nuclei

TX/RX Nucleus	1H
TX/RX Delta Frequency	0 Hz
TX Nucleus	None
TX Delta Frequency	0 Hz

Sequence - Special

Excite pulse duration	1000 us

Sequence - Special

Acq. window shift	200 us
Min. settling delay	300 us
Gradient ramp time	140 us
Use expt. SAR calc	Off
Check RF clipping	Off
Send ref. scans	Off
Inversion pulse	Off
Shift RO frequency	Off
Debug loop type	None

\\Research\Investigators\Frederick\CORPUS_CONSISTENT\eja_svs_press_diff
TA: 3:39 min Coil Selection: Auto Vol: 20×20×20 mm³ Rel. SNR: 1.00

Properties

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
Vol A >> P	20 mm
Vol R >> L	20 mm
Vol F >> H	20 mm
TR	3000.0 ms
TE	135.00 ms
Averages	64

Contrast - Common

TR	3000.0 ms
TE	135.00 ms
Excite flip angle	90 deg
Refocus flip angle	180 deg
Preparation Scans	4
Averages	64
VAPOR	Water Saturation
VAPOR suppr.	Water Suppr.
Water s. BW	135.00 Hz
Water s. Delta Pos.	0.00 ppm

Resolution - Common

Vector Size	2048
Normalize	Off

Geometry - Common

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
Vol A >> P	20 mm
Vol R >> L	20 mm
Vol F >> H	20 mm

Geometry - AutoAlign

AutoAlign	---
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Advanced
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off
Adj. Water Suppr.	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
A >> P	20 mm
R >> L	20 mm
F >> H	20 mm
Reset	Off

System - pTx

B1 Shim	TrueForm

System - Tx/Rx

Frequency 1H	123.249408 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	3000.0 ms

Sequence - Common

Sequence Name	press

Sequence - Common

Preparation Scans	4
Delta Frequency	-1.70 ppm
Measurements	1
Phase Cycling	Auto
Save Uncombined	Off
Bandwidth	2000 Hz
Acquisition Duration	1024 ms
Remove Oversampling	On

Sequence - Nuclei

TX/RX Nucleus	1H
TX/RX Delta Frequency	0 Hz
TX Nucleus	None
TX Delta Frequency	0 Hz

Sequence - Special

Excite pulse duration	2600 us
Refocus pulse duration	5200 us
Spoiler max. amplitude	20.00 mT/m
Spoiler amp. ratio	3.00
Refocus grad. factor	1.00
OVS slab thickness	80.00 mm
OVS slab pos. offset	5.00 mm
OVS delta frequency	0.00 ppm
Spoiler duration	1300 us
Acq. window shift	200 us
Min. settling delay	300 us
Gradient ramp time	140 us
Forced min. TE1	0 ms
VAPOR flip angle	80 deg
VAPOR delay 8	28 ms
VAPOR delay 7	74 ms
OVS pulse duration	5120 us
OVS HS pulse N	1
OVS HS pulse R	40
OVS flip angle RO	90 deg
OVS flip angle PH	90 deg
OVS flip angle SL	90 deg
VAPOR delay 6	68 ms
VAPOR delay 5	106 ms
VAPOR delay 4	105 ms
VAPOR delay 3	146 ms
VAPOR delay 2	100 ms
VAPOR delay 1	150 ms
Diffusion weighting	Off
Enable OVS	On
Resolve averages	Off
Use expt. SAR calc	Off
Check RF clipping	Off
Send ref. scans	Off
Inversion pulse	Off
'RR' refoc. pulse	Off
Invert SS grad. pol.	Off
Shift RO frequency	Off

Sequence - Special

Debug loop type	None

\\Research\Investigators\Frederick\CORPUS_CONSISTENT\eja_svs_slaser_diff
TA: 3:39 min Coil Selection: Auto Vol: 20×20×20 mm³ Rel. SNR: 1.00

Properties

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
Vol A >> P	20 mm
Vol R >> L	20 mm
Vol F >> H	20 mm
TR	3000.0 ms
TE	135.00 ms
Averages	64

Contrast - Common

TR	3000.0 ms
TE	135.00 ms
Excite flip angle	90 deg
Refocus flip angle	180 deg
Preparation Scans	4
Averages	64
VAPOR	Water Saturation
VAPOR suppr.	Water Suppr.
Water s. BW	135.00 Hz
Water s. Delta Pos.	0.00 ppm

Resolution - Common

Vector Size	2048
Normalize	Off

Geometry - Common

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
Vol A >> P	20 mm
Vol R >> L	20 mm
Vol F >> H	20 mm

Geometry - AutoAlign

AutoAlign	---
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Advanced
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off
Adj. Water Suppr.	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
A >> P	20 mm
R >> L	20 mm
F >> H	20 mm
Reset	Off

System - pTx

B1 Shim	TrueForm

System - Tx/Rx

Frequency 1H	123.249408 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	3000.0 ms

Sequence - Common

Sequence Name	slasr

Sequence - Common

Preparation Scans	4
Delta Frequency	-1.70 ppm
Measurements	1
Phase Cycling	Auto
Save Uncombined	Off
Bandwidth	2000 Hz
Acquisition Duration	1024 ms
Remove Oversampling	On

Sequence - Nuclei

TX/RX Nucleus	1H
TX/RX Delta Frequency	0 Hz
TX Nucleus	None
TX Delta Frequency	0 Hz

Sequence - Special

Excite pulse duration	8960 us
Refocus pulse duration	8960 us
Spoiler max. amplitude	20.00 mT/m
Spoiler amp. ratio	3.00
Refocus grad. factor	1.00
OVS slab thickness	80.00 mm
OVS slab pos. offset	5.00 mm
OVS delta frequency	0.00 ppm
Spoiler duration	1300 us
Acq. window shift	200 us
Min. settling delay	300 us
Gradient ramp time	140 us
HS refoc. pulse N	1
HS refoc. pulse R	20
VAPOR flip angle	80 deg
VAPOR delay 8	28 ms
VAPOR delay 7	74 ms
OVS pulse duration	5120 us
OVS HS pulse N	1
OVS HS pulse R	40
OVS flip angle RO	90 deg
OVS flip angle PH	0 deg
OVS flip angle SL	0 deg
VAPOR delay 6	68 ms
VAPOR delay 5	106 ms
VAPOR delay 4	105 ms
VAPOR delay 3	146 ms
VAPOR delay 2	100 ms
VAPOR delay 1	150 ms
Diffusion weighting	Off
Enable OVS	On
Resolve averages	Off
Use expt. SAR calc	Off
Check RF clipping	Off
Send ref. scans	Off
Inversion pulse	Off
GOIA refoc. pulses	Off
Asym. excit. pulse	Off

Sequence - Special

Invert SS grad. pol.	Off
Shift RO frequency	Off
Debug loop type	None

\\Research\Investigators\Frederick\CORPUS_CONSISTENT\ep_moco_nav_set_ABCD
TA: 1 sec Coil Selection: Auto Voxel Size: 8.0×8.0×4.0 mm³ Rel. SNR: 1.00

Properties

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slab Group	1
Slabs	1
Distance Factor	50 %
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slices per Slab	32
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	256 mm
FOV Phase	100.0 %
Slice Thickness	4.00 mm
TR	13.0 ms
TE	6.10 ms
Averages	1
Concatenations	1
AutoAlign	---

Contrast - Common

TR	13.0 ms
TE	6.10 ms
MTC	Off
Flip Angle	2 deg
Fat-Water Contrast	Standard
Contrasts	1
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Multiple Series	Each Measurement

Resolution - Common

FOV Read	256 mm

Resolution - Common

FOV Phase	100.0 %
Slice Thickness	4.00 mm
Base Resolution	32
Phase Resolution	100 %
Slice Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Phase Partial Fourier	Off
Slice Partial Fourier	6/8

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Hamming	Off
Distortion Correction	Off
Normalize	Off

Geometry - Common

Slab Group	1
Slabs	1
Distance Factor	50 %
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slices per Slab	32
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	256 mm
FOV Phase	100.0 %
Slice Thickness	4.00 mm
TR	13.0 ms
Multi-Slice Mode	Interleaved
Series	Interleaved
Concatenations	1

Geometry - AutoAlign

Slab Group	1
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
AutoAlign	---
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Sagittal
Initial Rotation	0.00 deg

Geometry - Saturation

Special Saturation	None

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Sum of Squares
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Tune up
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
A >> P	263 mm
R >> L	350 mm
F >> H	350 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Non-sel.

System - Tx/Rx

Frequency 1H	123.249408 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	13.0 ms
Concatenations	1

Sequence - Part 1

Sequence Name	ABCD

Sequence - Part 1

Dimension	3D
Excitation	Non-sel.
RF Pulse Type	Normal
Gradient Mode	Fast
Flow Compensation	None
Bandwidth	4882 Hz/Px
Echo Spacing	0.27 ms
Free Echo Spacing	Off
EPI Factor	32

Sequence - Part 2

Introduction	Off
RF Spoiling	On

Sequence - Special

Protocol filename	Generic

Sequence - Assistant

SAR Assistant	Off

\\Research\Investigators\Frederick\CORPUS_CONSISTENT\ep_seg_se
TA: 7:06 min Coil Selection: Auto Voxel Size: 0.9×0.9×3.5 mm³ Rel. SNR: 1.00

Properties

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	On
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slice Group	1
Slices	29
Distance Factor	14 %
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	P >> A
Phase Oversampling	0 %
FOV Read	224 mm
FOV Phase	100.0 %
Slice Thickness	3.5 mm
TR	17760.0 ms
TE	91.00 ms
Averages	1
Concatenations	4
AutoAlign	Head > Brain

Contrast - Common

TR	17760.0 ms
TE	91.00 ms
TD	0.00 ms
MTC	Off
Magn. Preparation	Slice-sel. IR
TI	2900 ms
Flip Angle	90 deg
Fat-Water Contrast	Fat Saturation
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Multiple Series	Each Measurement

Resolution - Common

FOV Read	224 mm
FOV Phase	100.0 %
Slice Thickness	3.5 mm
Base Resolution	256
Phase Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Phase Partial Fourier	6/8

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Hamming	Off
Distortion Correction	2D
Normalize	Off

Geometry - Common

Slice Group	1
Slices	29
Distance Factor	14 %
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	P >> A
Phase Oversampling	0 %
FOV Read	224 mm
FOV Phase	100.0 %
Slice Thickness	3.5 mm
TR	17760.0 ms
Multi-Slice Mode	Interleaved
Series	Interleaved
Concatenations	4

Geometry - AutoAlign

Slice Group	1
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	P >> A
AutoAlign	Head > Brain
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

Geometry - Saturation

Special Saturation	None

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off

Geometry - Tim Planning Suite

Table Position	0 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Standard
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	-180.00 deg
A >> P	224 mm
R >> L	224 mm
F >> H	116 mm
Reset	Off

System - pTx

B1 Shim	TrueForm

System - Tx/Rx

Frequency 1H	123.249408 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	17760.0 ms
Concatenations	4

Sequence - Part 1

Sequence Name	epir
Dimension	2D
RF Pulse Type	Normal
Gradient Mode	Fast
Bandwidth	652 Hz/Px
Echo Spacing	1.64 ms

Sequence - Part 1

Free Echo Spacing	Off
EPI Factor	65

Sequence - Part 2

Introduction	Off

Sequence - Assistant

SAR Assistant	Off

\\Research\Investigators\Frederick\CORPUS_CONSISTENT\fastestmap
TA: 8 sec Coil Selection: Manual Vol: 50×28×50 mm³ Rel. SNR: 1.00

Properties

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Position	L7.4 A35.1 H17.8 mm
Orientation	C > T-17.0
Rotation	0.00 deg
Vol R >> L	50 mm
Vol F >> H	28 mm
Vol A >> P	50 mm
TR	2800.0 ms
TE	50.00 ms
Averages	1
Coil Elements	HC3-6

Contrast - Common

TR	2800.0 ms
TE	50.00 ms
Tau	5.00 ms
Excite flip angle	90 deg
Refocus flip angle	180 deg
Averages	1
Water Suppression	None

Resolution - Common

Vector Size	256
Normalize	Off

Geometry - Common

Position	L7.4 A35.1 H17.8 mm
Orientation	C > T-17.0
Rotation	0.00 deg
Vol R >> L	50 mm
Vol F >> H	28 mm
Vol A >> P	50 mm

Geometry - AutoAlign

AutoAlign	---

Geometry - AutoAlign

Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

System - Miscellaneous

Coil Selection	Manual
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off
Adj. Water Suppr.	Off

System - Adjust Volume

Position	L7.4 A35.1 H17.8 mm
Orientation	C > T-17.0
Rotation	90.00 deg
F >> H	28 mm
R >> L	50 mm
A >> P	50 mm
Reset	Off

System - pTx

B1 Shim	TrueForm

System - Tx/Rx

Frequency 1H	123.249408 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Correction Factor	1.00
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	2800.0 ms

Sequence - Common

Sequence Name	fastmp

Sequence - Common

Delta Frequency	0.00 ppm
Measurements	1
Phase Cycling	None
Save Uncombined	Off
Bandwidth	100000 Hz
Acquisition Duration	2 ms

Sequence - Special

Type of fit	Linear 3-proj
VoI fit factor	100.00 %
Force spherical fit VoI	Off Force spherical fit VoI
Save plots to database	Off
Refocus pulses	Normal
Excite pulse duration	6400 us
Refocus pulse duration	6400 us
Bar FoV	384 mm
Bar thickness	10.00 mm
Inversion pulse	Off
Multi-echo acquisition	On
Number of echoes	4

\\Research\Investigators\Frederick\CORPUS_CONSISTENT\fid
TA: 1:08 min Coil Selection: Auto Rel. SNR: 1.00

Properties

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	Off
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

TR	1000.0 ms
TE	0.23 ms
Averages	64

Contrast - Common

TR	1000.0 ms
TE	0.23 ms
Pulse Type	Rectangular
Pulse Duration	0.26 ms
Flip Angle	50 deg
Preparation Scans	4
Averages	64
Water Suppression	None

Resolution - Common

Vector Size	1024
Normalize	Off

Geometry - AutoAlign

Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	When Changed
Assume Silicone	Off
Adj. Water Suppr.	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
A >> P	500 mm
R >> L	500 mm
F >> H	500 mm
Reset	Off

System - pTx

B1 Shim	TrueForm

System - Tx/Rx

Frequency 31P	49.891876 MHz
Frequency 1H	123.249408 MHz
? Ref. Amplitude 31P	0.000 V
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	1000.0 ms

Sequence - Common

Sequence Name	fid
Preparation Scans	4
Measurements	1
Phase Cycling	Two Step
Save Uncombined	Off
Acquisition Delay	0.10 ms
Bandwidth	2000 Hz
Acquisition Duration	512 ms
Remove Oversampling	On

Sequence - Nuclei

TX/RX Nucleus	31P
TX/RX Delta Frequency	0 Hz
NOE Type	None
TX Nucleus	1H
TX Delta Frequency	0 Hz

Sequence - Nuclei

Decoupling Type	None

\\Research\Investigators\Frederick\CORPUS_CONSISTENT\fl3d_rd
TA: 25 sec Coil Selection: Auto Voxel Size: 1.2×1.2×5.0 mm³ Rel. SNR: 1.00

Properties

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	On
Load Images to Graphic Segments	On
Graphic segment	2nd Segment
Inline Movie	Off

Routine

Slab Group	1
Slabs	1
Distance Factor	50 %
Position	L10.0 P0.0 F90.0 mm
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slices per Slab	26
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	300 mm
FOV Phase	59.4 %
Slice Thickness	5.00 mm
TR	23.0 ms
TE	8.00 ms
Averages	1
Concatenations	1
AutoAlign	Head > Brain

Contrast - Common

TR	23.0 ms
TE	8.00 ms
MTC	Off
Flip Angle	10 deg
Fat-Water Contrast	Standard
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Multiple Series	Off

Resolution - Common

FOV Read	300 mm
FOV Phase	59.4 %

Resolution - Common

Slice Thickness	5.00 mm
Base Resolution	256
Phase Resolution	75 %
Slice Resolution	54 %
Interpolation	Off

Resolution - Acceleration

Phase Partial Fourier	6/8
Slice Partial Fourier	6/8
Elliptical Scanning	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	2D
Normalize	Off
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	1
Distance Factor	50 %
Position	L10.0 P0.0 F90.0 mm
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slices per Slab	26
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	300 mm
FOV Phase	59.4 %
Slice Thickness	5.00 mm
TR	23.0 ms
Multi-Slice Mode	Interleaved
Series	Interleaved
Concatenations	1

Geometry - AutoAlign

Slab Group	1
Position	L10.0 P0.0 F90.0 mm
Orientation	Sagittal
Phase Encoding Dir.	A >> P
AutoAlign	Head > Brain
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

Geometry - Saturation

Special Saturation	None

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	C - T - S
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Sum of Squares
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Tune up
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
A >> P	263 mm
R >> L	350 mm
F >> H	350 mm
Reset	Off

System - pTx

B1 Shim	TrueForm

System - Tx/Rx

Frequency 1H	123.249408 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	Off

Inline - MIP

MIP Sag	On
MIP Cor	Off
MIP Tra	Off
MIP Time	Off
Radial MIP	Off
Save Original Images	Off
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Composing

Inline Composing	Off

Sequence - Part 1

Sequence Name	fl_rd
Dimension	3D
RF Pulse Type	Normal
Gradient Mode	Normal
Flow Compensation	On
Bandwidth	300 Hz/Px

Sequence - Part 2

Introduction	On
RF Spoiling	On

Sequence - Assistant

SAR Assistant	Off

\\Research\Investigators\Frederick\CORPUS_CONSISTENT\fl3d_vibe
TA: 4:02 min Coil Selection: Auto Voxel Size: 0.2×0.2×3.0 mm³ Acc:: None Rel. SNR: 1.00

Properties

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	On
Load Images to Graphic Segments	On
Graphic segment	Default
Inline Movie	Off

Routine

Slab Group	1
Slabs	1
Distance Factor	20 %
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slices per Slab	22
Phase Oversampling	0 %
Slice Oversampling	45.5 %
FOV Read	160 mm
FOV Phase	100.0 %
Slice Thickness	3.0 mm
TR	15.00 ms
TE	2.60 ms
Averages	1
Concatenations	1
AutoAlign	---

Contrast - Common

TR	15.00 ms
TE	2.60 ms
Flip Angle	5 deg
Fat-Water Contrast	Standard
Contrasts	1
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Temporal Interpolation	1
Measurements	2
Pause after Meas. 1	0.0 s
Multiple Series	Each Measurement
3D Reordering	Standard
Time to Center	48.9 s
Burn Time to Center	Off

Resolution - Common

FOV Read	160 mm
FOV Phase	100.0 %
Slice Thickness	3.0 mm
Base Resolution	384
Phase Resolution	100 %
Slice Resolution	81 %
Trajectory	Cartesian
Interpolation	On

Resolution - Acceleration

Acceleration Mode	None
Phase Partial Fourier	Off
Slice Partial Fourier	6/8
Asymmetric Echo	Weak
Elliptical Scanning	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
POCS	Off
Distortion Correction	2D
Normalize	Off
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	1
Distance Factor	20 %
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slices per Slab	22
Phase Oversampling	0 %
Slice Oversampling	45.5 %
FOV Read	160 mm
FOV Phase	100.0 %
Slice Thickness	3.0 mm
TR	15.00 ms
Series	Ascending
Concatenations	1

Geometry - AutoAlign

Slab Group	1
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
AutoAlign	---
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm

Geometry - AutoAlign

H	0.0 mm
Initial Orientation	Sagittal
Initial Rotation	0.00 deg

Geometry - Saturation

Special Saturation	None

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	ACS Restricted
Radial Sorting	Off
MSMA	S - C - T
Sagittal	Med >> Lat
Coronal	A >> P
Transversal	H >> F
Coil Combination	Adaptive Combine
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Tune up
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
A >> P	263 mm
R >> L	350 mm
F >> H	350 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Slab-sel.

System - Tx/Rx

Frequency 1H	123.249408 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - PACE

Resp. Control	Off
Concatenations	1

Inline - Liver

Liver Registration	Off
Save Original Images	On

Inline - Subtraction

Subtract	Off
Measurements	2
StdDev	Off
Save Original Images	On

Inline - MIP

MIP Sag	Off
MIP Cor	Off
MIP Tra	Off
MIP Time	Off
Radial MIP	Off
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Soft Tissue

Wash-in	Off
Wash-out	Off
TTP	Off
PEI	Off
MIP Time	Off
Measurements	2
Pause after Meas. 1	0.0 s

Inline - Composing

Inline Composing	Off

Sequence - Part 1

Sequence Name	fl
Dimension	3D
Excitation	Slab-sel.
RF Pulse Type	Normal
Gradient Mode	Fast
Bandwidth	280 Hz/Px
Asymmetric Echo	Weak
Optimization	Min. TE

Sequence - Part 2

Introduction	Off
RF Spoiling	On
Incr. Gradient Spoiling	Off
Breast Application	Off
Phase Enc. Order	Automatic

Sequence - Assistant

SAR Assistant	Off

Sequence - Assistant

Allowed Delay	0 s
Optimization	Min. TE

\\Research\Investigators\Frederick\CORPUS_CONSISTENT\fl_pc
TA: 22 sec Coil Selection: Auto Voxel Size: 1.2×1.2×5.0 mm³ Acc:: 3 Rel. SNR: 1.00

Properties

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	On
Load Images to Graphic Segments	On
Graphic segment	3rd Segment
Inline Movie	Off

Routine

Slab Group	1
Slabs	1
Distance Factor	20 %
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slices per Slab	24
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	300 mm
FOV Phase	68.8 %
Slice Thickness	5.00 mm
TR	32.3 ms
TE	5.89 ms
Averages	1
Concatenations	1
AutoAlign	---

Contrast - Common

TR	32.3 ms
TE	5.89 ms
Flip Angle	10 deg
Contrasts	1
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Multiple Series	Each Measurement

Resolution - Common

FOV Read	300 mm
FOV Phase	68.8 %
Slice Thickness	5.00 mm

Resolution - Common

Base Resolution	256
Phase Resolution	70 %
Slice Resolution	50 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	GRAPPA
Reference Scans	Integrated
Acceleration Factor PE	3
Reference Lines PE	24
Acceleration Factor 3D	1
Phase Partial Fourier	Off
Asymmetric Echo	Weak
Elliptical Scanning	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	On
Distortion Correction	2D
Normalize	Prescan
Image Filter	On

Geometry - Common

Slab Group	1
Slabs	1
Distance Factor	20 %
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slices per Slab	24
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	300 mm
FOV Phase	68.8 %
Slice Thickness	5.00 mm
TR	32.3 ms
Multi-Slice Mode	Sequential
Series	Interleaved
Concatenations	1

Geometry - AutoAlign

Slab Group	1
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
AutoAlign	---
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm

Geometry - AutoAlign

Initial Orientation	Sagittal
Initial Rotation	0.00 deg

Geometry - Saturation

Special Saturation	None

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Tune up
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
A >> P	263 mm
R >> L	350 mm
F >> H	350 mm
Reset	Off

System - pTx

B1 Shim	TrueForm

System - Tx/Rx

Frequency 1H	123.249408 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	32.3 ms
Segments	1

Physio - Signal

Concatenations	1

Flow

Flow Mode	Free
Encodings	2
Velocity Enc. 1	30 cm/s
Velocity Enc. 2	30 cm/s
Direction 1	F >> H
Direction 2	Through Plane
Rephased Images	On
Magnitude Images	Off
Magnitude Sum	On
Phase Images	Off

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - MIP

MIP Sag	On
MIP Cor	On
MIP Tra	Off
MIP Time	Off
Radial MIP	Off
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Composing

Inline Composing	Off

Sequence - Part 1

Sequence Name	pc
Dimension	3D
RF Pulse Type	Fast
Gradient Mode	Normal
Flow Compensation	None
Bandwidth	349 Hz/Px
Asymmetric Echo	Weak
Segments	1

Sequence - Part 2

Introduction	Off
RF Spoiling	On

Sequence - Assistant

SAR Assistant	Off

\\Research\Investigators\Frederick\CORPUS_CONSISTENT\gre
TA: 19 sec Coil Selection: Auto Voxel Size: 0.5×0.5×10.0 mm³ Acc:: None Rel. SNR: 1.00

Properties

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	On
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slice Group	1
Slices	5
Distance Factor	20 %
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slice Group	2
Slices	5
Distance Factor	20 %
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slice Group	3
Slices	5
Distance Factor	20 %
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FOV Read	280 mm
FOV Phase	100.0 %
Slice Thickness	10.0 mm
TR	20.0 ms
TE	5.00 ms
Averages	1
Concatenations	7
AutoAlign	---

Contrast - Common

TR	20.0 ms
TE	5.00 ms
TD	0.00 ms
MTC	Off
Magn. Preparation	None
Flip Angle	40 deg

Contrast - Common

Fat-Water Contrast	Standard
Dark Blood	Off
Contrasts	1
SWI	Off
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Multiple Series	Each Measurement

Resolution - Common

FOV Read	280 mm
FOV Phase	100.0 %
Slice Thickness	10.0 mm
Base Resolution	256
Phase Resolution	50 %
Interpolation	On

Resolution - Acceleration

Acceleration Mode	None
Deep Resolve	Off
Phase Partial Fourier	Off
Asymmetric Echo	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	On
Distortion Correction	2D
Normalize	Image Based
Image Filter	Off

Geometry - Common

Slice Group	1
Slices	5
Distance Factor	20 %
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slice Group	2
Slices	5
Distance Factor	20 %
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slice Group	3
Slices	5
Distance Factor	20 %
Position	Isocenter
Orientation	Sagittal

Geometry - Common

Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FOV Read	280 mm
FOV Phase	100.0 %
Slice Thickness	10.0 mm
TR	20.0 ms
Multi-Slice Mode	Sequential
Series	Interleaved
Concatenations	7

Geometry - AutoAlign

Slice Group	1
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slice Group	2
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slice Group	3
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
AutoAlign	---
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Sagittal
Initial Rotation	0.00 deg

Geometry - Saturation

Saturation Mode	Standard
Special Saturation	None

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	ACS All but spine
Radial Sorting	Off
MSMA	S - C - T
Sagittal	L >> R
Coronal	P >> A
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard

System - Adjustments

B0 Shim	Tune up
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	On
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
A >> P	263 mm
R >> L	350 mm
F >> H	350 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Slice-sel.
LR Balancing	Off

System - Tx/Rx

Frequency 1H	123.249408 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	20.0 ms
Segments	1
Concatenations	7

Physio - Cardiac

Fat-Water Contrast	Standard
Magn. Preparation	None
Dark Blood	Off
FOV Read	280 mm
FOV Phase	100.0 %
Phase Resolution	50 %

Physio - PACE

Resp. Control	Off
Concatenations	7

Inline - Liver

Liver Registration	Off
Save Original Images	On

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - MIP

MIP Sag	Off
MIP Cor	Off
MIP Tra	Off
MIP Time	Off
Radial MIP	Off
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Soft Tissue

Wash-in	Off
Wash-out	Off
TTP	Off
PEI	Off
MIP Time	Off
Measurements	1

Inline - Composing

Inline Composing	Off

Sequence - Part 1

Sequence Name	fl
Dimension	2D
Excitation	Slice-sel.
RF Pulse Type	Fast
Gradient Mode	Normal
Flow Compensation	None
Bandwidth	180 Hz/Px
Asymmetric Echo	Off
Segments	1

Sequence - Part 2

Introduction	On
RF Spoiling	On
Acoustic noise reduction	Off

Sequence - Assistant

SAR Assistant	Off
Allowed Delay	0 s

\\Research\Investigators\Frederick\CORPUS_CONSISTENT\gre_field_mapping
TA: 1:30 min Coil Selection: Auto Voxel Size: 3.4×3.4×2.0 mm³ Rel. SNR: 1.00

Properties

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slice Group	1
Slices	72
Distance Factor	0 %
Position	Isocenter
Orientation	T > C-22.0
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FOV Read	220 mm
FOV Phase	96.9 %
Slice Thickness	2.0 mm
TR	700.0 ms
TE 1	4.92 ms
TE 2	7.38 ms
Averages	1
Concatenations	1
AutoAlign	---

Contrast - Common

TR	700.0 ms
TE 1	4.92 ms
TE 2	7.38 ms
MTC	Off
Flip Angle	60 deg
Fat-Water Contrast	Standard
Contrasts	2
Reconstruction	Magn./Phase

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Multiple Series	Off

Resolution - Common

FOV Read	220 mm
FOV Phase	96.9 %
Slice Thickness	2.0 mm
Base Resolution	64
Phase Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Phase Partial Fourier	Off
Asymmetric Echo	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	2D
Normalize	Prescan
Image Filter	Off

Geometry - Common

Slice Group	1
Slices	72
Distance Factor	0 %
Position	Isocenter
Orientation	T > C-22.0
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FOV Read	220 mm
FOV Phase	96.9 %
Slice Thickness	2.0 mm
TR	700.0 ms
Multi-Slice Mode	Interleaved
Series	Interleaved
Concatenations	1

Geometry - AutoAlign

Slice Group	1
Position	Isocenter
Orientation	T > C-22.0
Phase Encoding Dir.	A >> P
AutoAlign	---
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	T > C
T > C	-22.00
> S	0.00
Initial Rotation	0.00 deg

Geometry - Saturation

Special Saturation	None

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	ACS All but spine
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Sum of Squares
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Standard
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	On
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	Isocenter
Orientation	T > C-22.0
Rotation	0.00 deg
A >> P	214 mm
R >> L	220 mm
F >> H	144 mm
Reset	Off

System - pTx

B1 Shim	TrueForm

System - Tx/Rx

Frequency 1H	123.249408 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Sequence - Part 1

Sequence Name	fm_r
Dimension	2D
RF Pulse Type	Normal
Gradient Mode	Fast
Flow Compensation	On
Bandwidth	596 Hz/Px
Asymmetric Echo	Off

Sequence - Part 2

Introduction	On

Sequence - Part 2

RF Spoiling	On

Sequence - Assistant

SAR Assistant	Off

\\Research\Investigators\Frederick\CORPUS_CONSISTENT\haste
TA: 9 sec Coil Selection: Auto Voxel Size: 0.9×0.9×4.0 mm³ Acc:: 2 Rel. SNR: 1.00

Properties

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	On
Load Images to Graphic Segments	On
Graphic segment	Default
Inline Movie	Off

Routine

Slice Group	1
Slices	5
Distance Factor	30 %
Position	L0.0 A25.0 H0.0 mm
Orientation	Transversal
Phase Encoding Dir.	R >> L
Phase Oversampling	0 %
FOV Read	240 mm
FOV Phase	100.0 %
Slice Thickness	4.0 mm
TR	1500.0 ms
TE	58.00 ms
Averages	1
Concatenations	1
AutoAlign	---

Contrast - Common

TR	1500.0 ms
TE	58.00 ms
MTC	Off
Magn. Preparation	None
Flip Angle Mode	Constant
Flip Angle	150 deg
Fat-Water Contrast	Standard
Dark Blood	Off
Contrasts	1
Wrap-up Magn.	None
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Multiple Series	Each Measurement

Resolution - Common

FOV Read	240 mm
FOV Phase	100.0 %
Slice Thickness	4.0 mm
Base Resolution	256
Phase Resolution	80 %
Interpolation	1.00

Resolution - Acceleration

Acceleration Mode	GRAPPA
Reference Scans	Integrated
Acceleration Factor PE	2
Reference Lines PE	24
Deep Resolve	Off
Phase Partial Fourier	4/8

Resolution - Filter

Raw Filter	Off
Elliptical Filter	On
Distortion Correction	2D
Normalize	Prescan
Image Filter	Off

Geometry - Common

Slice Group	1
Slices	5
Distance Factor	30 %
Position	L0.0 A25.0 H0.0 mm
Orientation	Transversal
Phase Encoding Dir.	R >> L
Phase Oversampling	0 %
FOV Read	240 mm
FOV Phase	100.0 %
Slice Thickness	4.0 mm
TR	1500.0 ms
Multi-Slice Mode	Single Shot
Series	Interleaved
Concatenations	1

Geometry - AutoAlign

Slice Group	1
Position	L0.0 A25.0 H0.0 mm
Orientation	Transversal
Phase Encoding Dir.	R >> L
AutoAlign	---
Initial Position	L0.0 A25.0 H0.0
L	0.0 mm
A	25.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	90.01 deg

Geometry - Navigator

Geometry - Saturation

Special Saturation	None

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	ACS All but spine
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Tune up
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
A >> P	263 mm
R >> L	350 mm
F >> H	350 mm
Reset	Off

System - pTx

B1 Shim	TrueForm

System - Tx/Rx

Frequency 1H	123.249408 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	1500.0 ms
Concatenations	1

Physio - Cardiac

Fat-Water Contrast	Standard
Magn. Preparation	None
Dark Blood	Off
FOV Read	240 mm
FOV Phase	100.0 %
Phase Resolution	80 %
Motion Correction	None

Physio - PACE

Resp. Control	Off
Concatenations	1

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Motion Correction	None
Save Original Images	On

Inline - MIP

MIP Sag	Off
MIP Cor	Off
MIP Tra	Off
MIP Time	Off
Radial MIP	Off
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Composing

Inline Composing	Off

Sequence - Part 1

Sequence Name	h
Dimension	2D
RF Pulse Type	Fast
Gradient Mode	Fast
Flow Compensation	None
Bandwidth	592 Hz/Px
Echo Spacing	4.46 ms
Turbo Factor	205

Sequence - Part 2

Introduction	On
Motion Correction	None

Sequence - Assistant

SAR Assistant	Off
Allowed Delay	30 s

\\Research\Investigators\Frederick\CORPUS_CONSISTENT\se_mc

TA: 6:35 min Coil Selection: Auto Voxel Size: 0.4×0.4×3.0 mm³ Acc:: None Rel. SNR: 1.00

Properties

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	On
Load Images to Graphic Segments	On
Graphic segment	Default
Inline Movie	Off

Routine

Slice Group	1
Slices	25
Distance Factor	20 %
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FOV Read	160 mm
FOV Phase	100.0 %
Slice Thickness	3.0 mm
TR	1910.0 ms
TE 1	13.80 ms
TE 2	27.60 ms
TE 3	41.40 ms
TE 4	55.20 ms
TE 5	69.00 ms
Averages	1
Concatenations	1
AutoAlign	---

Contrast - Common

TR	1910.0 ms
TE 1	13.80 ms
TE 2	27.60 ms
TE 3	41.40 ms
TE 4	55.20 ms
TE 5	69.00 ms
MTC	Off
Magn. Preparation	None
Flip Angle	180 deg
Fat-Water Contrast	Standard
Dark Blood	Off
Contrasts	5
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Multiple Series	Each Measurement

Resolution - Common

FOV Read	160 mm
FOV Phase	100.0 %
Slice Thickness	3.0 mm
Base Resolution	384
Phase Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	None
Phase Partial Fourier	4/8

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	2D
Normalize	Off
Image Filter	Off

Geometry - Common

Slice Group	1
Slices	25
Distance Factor	20 %
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FOV Read	160 mm
FOV Phase	100.0 %
Slice Thickness	3.0 mm
TR	1910.0 ms
Multi-Slice Mode	Interleaved
Series	Interleaved
Concatenations	1

Geometry - AutoAlign

Slice Group	1
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
AutoAlign	---
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Sagittal

Geometry - AutoAlign

Initial Rotation	0.00 deg

Geometry - Saturation

Special Saturation	None

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	ACS Restricted
Radial Sorting	Off
MSMA	S - C - T
Sagittal	Med >> Lat
Coronal	A >> P
Transversal	H >> F
Coil Combination	Adaptive Combine
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Tune up
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
A >> P	263 mm
R >> L	350 mm
F >> H	350 mm
Reset	Off

System - pTx

B1 Shim	TrueForm

System - Tx/Rx

Frequency 1H	123.249408 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	1910.0 ms
Concatenations	1

Physio - Cardiac

Fat-Water Contrast	Standard
Magn. Preparation	None
Dark Blood	Off
FOV Read	160 mm
FOV Phase	100.0 %
Phase Resolution	100 %

Inline - Liver

Liver Registration	Off
Save Original Images	On

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - MIP

MIP Sag	Off
MIP Cor	Off
MIP Tra	Off
MIP Time	Off
Radial MIP	Off
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Composing

Inline Composing	Off

Sequence - Part 1

Sequence Name	se
RF Pulse Type	Low SAR
Gradient Mode	Fast
Bandwidth	228 Hz/Px

Sequence - Part 2

Introduction	On

Sequence - Assistant

SAR Assistant	Off
Allowed Delay	0 s

\\Research\Investigators\Frederick\CORPUS_CONSISTENT\space
TA: 9:54 min Coil Selection: Auto Voxel Size: 1.0×1.0×1.0 mm³ Acc:: None Rel. SNR: 1.00

Properties

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	On
Load Images to Graphic Segments	On
Graphic segment	Default
Inline Movie	Off

Routine

Slab Group	1
Slabs	1
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slices per Slab	176
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	250 mm
FOV Phase	100.0 %
Slice Thickness	1.00 mm
TR	700.0 ms
TE	11.00 ms
Averages	1.0
Concatenations	1
AutoAlign	Head > Basis

Contrast - Common

TR	700.0 ms
TE	11.00 ms
MTC	Off
Magn. Preparation	None
Flip Angle Mode	T1 Var
Fat-Water Contrast	Standard
Dark Blood	Off
Blood Suppression	Off
Wrap-up Magn.	Restore
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Multiple Series	Each Measurement
Reordering	Linear

Resolution - Common

FOV Read	250 mm
FOV Phase	100.0 %
Slice Thickness	1.00 mm
Base Resolution	256
Phase Resolution	100 %
Slice Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	None
Phase Partial Fourier	Off
Slice Partial Fourier	Off
Elliptical Scanning	On

Resolution - Filter

Raw Filter	On
Elliptical Filter	Off
Distortion Correction	2D
Normalize	Prescan
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	1
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slices per Slab	176
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	250 mm
FOV Phase	100.0 %
Slice Thickness	1.00 mm
TR	700.0 ms
Concatenations	1

Geometry - AutoAlign

Slab Group	1
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
AutoAlign	Head > Basis
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Sagittal
Initial Rotation	0.00 deg

Geometry - Navigator

Geometry - Saturation

Special Saturation	None

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Performance
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Standard
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	Isocenter
Orientation	Sagittal
Rotation	0.00 deg
A >> P	250 mm
F >> H	250 mm
R >> L	176 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Non-sel.

System - Tx/Rx

Frequency 1H	123.249408 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000
Gain	High

Physio - Signal

1st Signal/Mode	None
Trigger Delay	0 ms
TR	700.0 ms
Concatenations	1

Physio - Cardiac

Fat-Water Contrast	Standard
Magn. Preparation	None
Dark Blood	Off
FOV Read	250 mm
FOV Phase	100.0 %
Phase Resolution	100 %

Physio - PACE

Resp. Control	Off
Concatenations	1

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - MIP

MIP Sag	Off
MIP Cor	Off
MIP Tra	Off
MIP Time	Off
Radial MIP	Off
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Composing

Inline Composing	Off

Sequence - Part 1

Sequence Name	spcR
Dimension	3D
Excitation	Non-sel.
RF Pulse Type	Fast
Gradient Mode	Fast
Flow Compensation	None
Reordering	Linear
Bandwidth	630 Hz/Px
Echo Spacing	3.72 ms
Turbo Factor	42
Echo Train Duration	167 ms

Sequence - Part 2

Introduction	On

Sequence - Assistant

SAR Assistant	Off
Allowed Delay	30 s

\\Research\Investigators\Frederick\CORPUS_CONSISTENT\ZPL_RG_EPSI_FID_v2e
TA: 12 sec Coil Selection: Auto Voxel Size: 1.1×1.0×1.7 mm³ Rel. SNR: 1.00

Properties

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
FOV A >> P	240 mm
FOV R >> L	240 mm
FOV F >> H	120 mm
Vol A >> P	240 mm
Vol R >> L	240 mm
Vol F >> H	100 mm
TR	67.0 ms
TE	1.60 ms
Averages	1

Contrast - Common

TR	67.0 ms
TE	1.60 ms
Ernst angle	18 deg
Prep. pulses	38 #
T1 Estimated	1400 ms
Flip Angle	32 deg
Preparation Scans	0
Averages	1
Water Suppression	None

Resolution - Common

FOV A >> P	240 mm
FOV R >> L	240 mm
FOV F >> H	120 mm
Scan Res. A >> P	216
Scan Res. R >> L	230
Scan Res. F >> H	72
Interpol. Res. A >> P	256
Interpol. Res. R >> L	256
Interpol. Res. F >> H	128
Hamming	Off

Resolution - Common

Dimension	3D
Phase Encoding	Full
Normalize	Off

Geometry - Common

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
FOV A >> P	240 mm
FOV R >> L	240 mm
FOV F >> H	120 mm
Vol A >> P	240 mm
Vol R >> L	240 mm
Vol F >> H	100 mm
Fully Excited Vol	Off
Saturation Region	1
Thickness	100.00 mm
Position	L0.0 P0.0 H104.7 mm
Orientation	Transversal
Sat. Delta Frequ.	0.00 ppm
Saturation Region	2
Thickness	100.00 mm
Position	L0.0 P0.0 H104.7 mm
Orientation	Transversal
Sat. Delta Frequ.	0.00 ppm

Geometry - AutoAlign

Slab Group	1
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
AutoAlign	---
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Advanced

System - Adjustments

B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off
Adj. Water Suppr.	Off

System - Adjust Volume

! Position	Isocenter
! Orientation	Transversal
! Rotation	0.00 deg
! A >> P	240 mm
! R >> L	240 mm
! F >> H	100 mm
Reset	Off

System - pTx

B1 Shim	TrueForm

System - Tx/Rx

Frequency 1H	123.249408 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Sequence - Common

Sequence Name	ZPL_epsi_fid
Preparation Scans	0
Delta Frequency	0.00 ppm
Dimension	3D
Save Uncombined	Off
Acquisition Delay	2.30 ms
Bandwidth	167000 Hz
Acquisition Duration	0 ms
Remove Oversampling	Off

Sequence - Nuclei

TX/RX Nucleus	1H
TX/RX Delta Frequency	0 Hz
TX Nucleus	None
TX Delta Frequency	0 Hz

Sequence - Special

PE Samp EPSI	SegsV2g_216x72_COy4Rz1SEGz2Prep9.dat
PE Samp Prep	No Customized Sampling Pattern
Mode: T2-Prep	9 #
Mode: MotionNav	0 #
Mode: WatSupNav	0 #
Mode: BipolNav	1 #
Mode: RFspoil	0 #
Ramping time (PE)	180 us
Grad duration (PE)	510 us
Blip ramp time (PE)	60 us

Sequence - Special

Blip ramp time (SL)	60 us
Spoil duration (SP)	4000 us
EPSI. Num: Echo	18 #
EPSI. Echo: Shift	0 #
EPSI. BlipLeng: Phas	256 #
EPSI. BlipStep: Phas	3 #
EPSI. BlipStep: Slic	1 #
EPSI. Ramp. Samp	16 #
EPSI. RO. Ramp time	110 us
EPSI. EchoSpace	3200 us
T2Prep. Echo time	100000 us
T2Prep. Pulse mode	1 #
Prep. Num Wait TR	4 #
Pulse duration (RF)	1200 us
Pulse bandwidth (RF)	7291 Hz
FOV Pos, Read	0 um
FOV Pos, Phase	0 um
FOV Pos, Slice	0 um

\\Research\Investigators\Frederick\CORPUS_CONSISTENT\can_neuromelanin
TA: 14 sec Coil Selection: Auto Voxel Size: 2.3×2.3×5.0 mm³ Acc:: None Rel. SNR: 1.00

Properties

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slice Group	1
Slices	1
Distance Factor	20 %
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FOV Read	300 mm
FOV Phase	100.0 %
Slice Thickness	5.0 mm
TR	100.0 ms
TE	10.00 ms
Averages	1
Concatenations	1
AutoAlign	---

Contrast - Common

TR	100.0 ms
TE	10.00 ms
MTC	Off
Magn. Preparation	None
Flip Angle	25 deg
Fat-Water Contrast	Standard
Dark Blood	Off
Contrasts	1
SWI	Off
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Multiple Series	Each Measurement

Resolution - Common

FOV Read	300 mm
FOV Phase	100.0 %
Slice Thickness	5.0 mm
Base Resolution	128
Phase Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	None
Phase Partial Fourier	Off
Asymmetric Echo	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	2D
Normalize	Off
Image Filter	Off

Geometry - Common

Slice Group	1
Slices	1
Distance Factor	20 %
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FOV Read	300 mm
FOV Phase	100.0 %
Slice Thickness	5.0 mm
TR	100.0 ms
Multi-Slice Mode	Interleaved
Series	Interleaved
Concatenations	1

Geometry - AutoAlign

Slice Group	1
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
AutoAlign	---
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

Geometry - Saturation

Saturation Mode	Standard

Geometry - Saturation

Special Saturation	None

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Tune up
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
A >> P	263 mm
R >> L	350 mm
F >> H	350 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Slice-sel.
LR Balancing	Off

System - Tx/Rx

Frequency 1H	123.249408 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	100.0 ms
Segments	1
Concatenations	1

Physio - Cardiac

Fat-Water Contrast	Standard
Magn. Preparation	None
Dark Blood	Off
FOV Read	300 mm
FOV Phase	100.0 %
Phase Resolution	100 %

Physio - PACE

Resp. Control	Off
Concatenations	1

Inline - Liver

Liver Registration	Off
Save Original Images	On

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - MIP

MIP Sag	Off
MIP Cor	Off
MIP Tra	Off
MIP Time	Off
Radial MIP	Off
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Soft Tissue

Wash-in	Off
Wash-out	Off
TTP	Off
PEI	Off
MIP Time	Off
Measurements	1

Inline - Composing

Inline Composing	Off

Sequence - Part 1

Sequence Name	fl
Dimension	2D
Excitation	Slice-sel.
RF Pulse Type	Normal
Gradient Mode	Fast
Flow Compensation	None
Bandwidth	260 Hz/Px
Asymmetric Echo	Off
Segments	1

Sequence - Part 2

Introduction	On
RF Spoiling	On
Acoustic noise reduction	Off

Sequence - Special

MT Flip Angle	370 degrees
MT Offset	1500 Hz

Sequence - Assistant

SAR Assistant	Off
Allowed Delay	0 s

\\Research\Investigators\Frederick\CORPUS_CONSISTENT\can_neuromelanin_pk
TA: 14 sec Coil Selection: Auto Voxel Size: 2.3×2.3×5.0 mm³ Acc:: None Rel. SNR: 1.00

Properties

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slice Group	1
Slices	1
Distance Factor	20 %
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FOV Read	300 mm
FOV Phase	100.0 %
Slice Thickness	5.0 mm
TR	100.0 ms
TE	10.00 ms
Averages	1
Concatenations	1
AutoAlign	---

Contrast - Common

TR	100.0 ms
TE	10.00 ms
MTC	Off
Magn. Preparation	None
Flip Angle	25 deg
Fat-Water Contrast	Standard
Dark Blood	Off
Contrasts	1
SWI	Off
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Multiple Series	Each Measurement

Resolution - Common

FOV Read	300 mm
FOV Phase	100.0 %
Slice Thickness	5.0 mm
Base Resolution	128
Phase Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	None
Phase Partial Fourier	Off
Asymmetric Echo	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	2D
Normalize	Off
Image Filter	Off

Geometry - Common

Slice Group	1
Slices	1
Distance Factor	20 %
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FOV Read	300 mm
FOV Phase	100.0 %
Slice Thickness	5.0 mm
TR	100.0 ms
Multi-Slice Mode	Interleaved
Series	Interleaved
Concatenations	1

Geometry - AutoAlign

Slice Group	1
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
AutoAlign	---
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

Geometry - Saturation

Saturation Mode	Standard

Geometry - Saturation

Special Saturation	None

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Tune up
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
A >> P	263 mm
R >> L	350 mm
F >> H	350 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Slice-sel.
LR Balancing	Off

System - Tx/Rx

Frequency 1H	123.249408 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	100.0 ms
Segments	1
Concatenations	1

Physio - Cardiac

Fat-Water Contrast	Standard
Magn. Preparation	None
Dark Blood	Off
FOV Read	300 mm
FOV Phase	100.0 %
Phase Resolution	100 %

Physio - PACE

Resp. Control	Off
Concatenations	1

Inline - Liver

Liver Registration	Off
Save Original Images	On

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - MIP

MIP Sag	Off
MIP Cor	Off
MIP Tra	Off
MIP Time	Off
Radial MIP	Off
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Soft Tissue

Wash-in	Off
Wash-out	Off
TTP	Off
PEI	Off
MIP Time	Off
Measurements	1

Inline - Composing

Inline Composing	Off

Sequence - Part 1

Sequence Name	fl
Dimension	2D
Excitation	Slice-sel.
RF Pulse Type	Normal
Gradient Mode	Fast
Flow Compensation	None
Bandwidth	260 Hz/Px
Asymmetric Echo	Off
Segments	1

Sequence - Part 2

Introduction	On
RF Spoiling	On
Acoustic noise reduction	Off

Sequence - Special

MT Flip Angle	370 degrees
MT Offset	1500 Hz
Apply Partial Kspace Coverage	Off

Sequence - Assistant

SAR Assistant	Off
Allowed Delay	0 s

\\Research\Investigators\Frederick\CORPUS_CONSISTENT\csi_se
TA: 7:56 min Coil Selection: Auto Voxel Size: 10.0×10.0×15.0 mm³ Rel. SNR: 1.00

Properties

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
FOV A >> P	160 mm
FOV R >> L	160 mm
Thickness F >> H	15 mm
Vol A >> P	80 mm
Vol R >> L	80 mm
TR	1700.0 ms
TE	30.00 ms
Averages	3

Contrast - Common

TR	1700.0 ms
TE	30.00 ms
Flip Angle	90 deg
Preparation Scans	4
Averages	3
Water Suppression	Water Saturation
Water Suppr. BW	50 Hz
Spectral Suppr.	None
Application	Body

Resolution - Common

FOV A >> P	160 mm
FOV R >> L	160 mm
Thickness F >> H	15 mm
Scan Res. A >> P	16
Scan Res. R >> L	16
Interpol. Res. A >> P	16
Interpol. Res. R >> L	16
Hamming	On
Dimension	2D
Phase Encoding	Weighted
Vector Size	1024

Resolution - Common

Normalize	Prescan

Geometry - Common

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
FOV A >> P	160 mm
FOV R >> L	160 mm
Thickness F >> H	15 mm
Vol A >> P	80 mm
Vol R >> L	80 mm
Fully Excited Vol	On
Saturation Region	1
Thickness	40.00 mm
Position	R62.0 P0.0 H0.0 mm
Orientation	Sagittal
Sat. Delta Frequ.	-3.60 ppm
Saturation Region	2
Thickness	40.00 mm
Position	R62.0 P0.0 H0.0 mm
Orientation	Sagittal
Sat. Delta Frequ.	-3.60 ppm
Saturation Region	3
Thickness	40.00 mm
Position	R62.0 P0.0 H0.0 mm
Orientation	Sagittal
Sat. Delta Frequ.	-3.60 ppm
Saturation Region	4
Thickness	40.00 mm
Position	R62.0 P0.0 H0.0 mm
Orientation	Sagittal
Sat. Delta Frequ.	-3.60 ppm

Geometry - AutoAlign

Slice Group	1
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
AutoAlign	---
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T

System - Miscellaneous

Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	When Changed
Assume Silicone	Off
Adj. Water Suppr.	On

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
A >> P	80 mm
R >> L	80 mm
F >> H	15 mm
Reset	Off

System - pTx

B1 Shim	TrueForm

System - Tx/Rx

Frequency 1H	123.249408 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Sequence - Common

Sequence Name	csi_se
Preparation Scans	4
Delta Frequency	-2.70 ppm
Measurements	1
Dimension	2D
Save Uncombined	Off
Bandwidth	1200 Hz
Acquisition Duration	853 ms
Remove Oversampling	On

\\Research\Investigators\Frederick\CORPUS_CONSISTENT\csi_slaser
TA: 7:56 min Coil Selection: Auto Voxel Size: 10.0×10.0×15.0 mm³ Rel. SNR: 1.00

Properties

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
FOV A >> P	160 mm
FOV R >> L	160 mm
Thickness F >> H	15 mm
Vol A >> P	80 mm
Vol R >> L	80 mm
TR	1700.0 ms
TE	40.00 ms
Averages	3

Contrast - Common

TR	1700.0 ms
TE	40.00 ms
Flip Angle	90 deg
Preparation Scans	4
Averages	3
Water Suppression	Water Saturation
Water Suppr. BW	50 Hz

Resolution - Common

FOV A >> P	160 mm
FOV R >> L	160 mm
Thickness F >> H	15 mm
Scan Res. A >> P	16
Scan Res. R >> L	16
Interpol. Res. A >> P	16
Interpol. Res. R >> L	16
Hamming	On
Dimension	2D
Phase Encoding	Weighted
Vector Size	1024
Normalize	Prescan

Geometry - Common

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
FOV A >> P	160 mm
FOV R >> L	160 mm
Thickness F >> H	15 mm
Vol A >> P	80 mm
Vol R >> L	80 mm
Saturation Region	1
Thickness	40.00 mm
Position	L61.0 P0.0 H0.0 mm
Orientation	Sagittal
Sat. Delta Frequ.	-3.40 ppm
Saturation Region	2
Thickness	40.00 mm
Position	L61.0 P0.0 H0.0 mm
Orientation	Sagittal
Sat. Delta Frequ.	-3.40 ppm
Saturation Region	3
Thickness	40.00 mm
Position	L61.0 P0.0 H0.0 mm
Orientation	Sagittal
Sat. Delta Frequ.	-3.40 ppm
Saturation Region	4
Thickness	40.00 mm
Position	L61.0 P0.0 H0.0 mm
Orientation	Sagittal
Sat. Delta Frequ.	-3.40 ppm

Geometry - AutoAlign

Slice Group	1
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
AutoAlign	---
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	When Changed
Assume Silicone	Off
Adj. Water Suppr.	On

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
A >> P	80 mm
R >> L	80 mm
F >> H	15 mm
Reset	Off

System - pTx

B1 Shim	TrueForm

System - Tx/Rx

Frequency 1H	123.249408 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Sequence - Common

Sequence Name	csislsr
Preparation Scans	4
Delta Frequency	-2.70 ppm
Measurements	1
Dimension	2D
Save Uncombined	Off
Bandwidth	1200 Hz
Acquisition Duration	853 ms
Remove Oversampling	On

\\Research\Investigators\Frederick\CORPUS_CONSISTENT\petra
TA: 4:41 min Coil Selection: Auto Voxel Size: 0.9×0.9×0.9 mm³ Rel. SNR: 1.00

Properties

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slab Group	1
Slabs	1
Position	Isocenter
Orientation	Transversal
Slices per Slab	320
FOV Read	300 mm
FOV Phase	100.0 %
Slice Thickness	0.94 mm
TR 1	3.32 ms
TR 2	2250.0 ms
TE	0.07 ms
AutoAlign	---

Contrast - Common

TR 1	3.32 ms
TR 2	2250.0 ms
TE	0.07 ms
Magn. Preparation	Non-sel. IR
TI 1	2000 ms
TI 2	900 ms
Flip Angle	6 deg
Fat-Water Contrast	Fat Saturation
Contrasts	1

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1

Resolution - Common

FOV Read	300 mm
FOV Phase	100.0 %
Slice Thickness	0.94 mm
Base Resolution	320
Radial Views	40000

Resolution - Acceleration

Phase Partial Fourier	Off

Resolution - Filter

Distortion Correction	2D
Normalize	Prescan
Image Filter	On

Geometry - Common

Slab Group	1
Slabs	1
Position	Isocenter
Orientation	Transversal
Slices per Slab	320
FOV Read	300 mm
FOV Phase	100.0 %
Slice Thickness	0.94 mm
TR 1	3.32 ms
TR 2	2250.0 ms

Geometry - AutoAlign

Slab Group	1
Position	Isocenter
Orientation	Transversal
AutoAlign	---
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

Geometry - Saturation

Special Saturation	None

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	ACS All but spine
Radial Sorting	Off
MSMA	S - C - T
Sagittal	L >> R
Coronal	P >> A
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Standard
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
A >> P	300 mm
R >> L	300 mm
F >> H	301 mm
Reset	Off

System - pTx

B1 Shim	TrueForm

System - Tx/Rx

Frequency 1H	123.249408 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Sequence - Part 1

Sequence Name	Petra
Dimension	3D
Gradient Mode	Whisper
Bandwidth	401 Hz/Px
Turbo Factor	400
Segments	20

Sequence - Part 2

Introduction	Off
Acoustic noise reduction	On

Sequence - Assistant

SAR Assistant	Off

\\Research\Investigators\Frederick\CORPUS_CONSISTENT\resolve
TA: 2:34 min Coil Selection: Auto Voxel Size: 1.1×1.1×4.0 mm³ Acc:: None Rel. SNR: 1.00

Properties

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slice Group	1
Slices	30
Distance Factor	25 %
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FOV Read	217 mm
FOV Phase	100.0 %
Slice Thickness	4.0 mm
TR	5870.0 ms
TE 1	82 ms
TE 2	151 ms
Concatenations	1
AutoAlign	Head > Brain

Contrast - Common

TR	5870.0 ms
TE 1	82 ms
TE 2	151 ms
MTC	Off
Magn. Preparation	None
Flip Angle	180 deg
Fat-Water Contrast	Fat Saturation
Fat Saturation	Strong
Contrasts	2
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1

Resolution - Common

FOV Read	217 mm

Resolution - Common

FOV Phase	100.0 %
Slice Thickness	4.0 mm
Base Resolution	192
Phase Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Accel. Mode	None
Phase Partial Fourier	Off
Readout Partial Fourier	5/8
Readout Segments	7

Resolution - Filter

Raw Filter	On
Distortion Correction	Off
Normalize	Prescan
Noise Masking	Off

Geometry - Common

Slice Group	1
Slices	30
Distance Factor	25 %
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FOV Read	217 mm
FOV Phase	100.0 %
Slice Thickness	4.0 mm
TR	5870.0 ms
Multi-Slice Mode	Interleaved
Series	Interleaved
Concatenations	1

Geometry - AutoAlign

Slice Group	1
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
AutoAlign	Head > Brain
Initial Position	L0.0 P0.0 F37.0
L	0.0 mm
P	0.0 mm
F	37.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

Geometry - Saturation

Special Saturation	None

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	ACS All but spine
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Standard
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
A >> P	217 mm
R >> L	217 mm
F >> H	149 mm
Reset	Off

System - pTx

B1 Shim	TrueForm

System - Tx/Rx

Frequency 1H	123.249408 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	5870.0 ms
Concatenations	1

Diff

Diffusion Mode	4-Scan Trace
Diff. Directions	4
Diffusion Scheme	Monopolar
Diff. Weightings	2
b-value 1	0 s/mm²

Diff

b-value 2	1000 s/mm²
Averages 1	1
Averages 2	1
Invert Gray Scale	Off
Diff. Weighted Images	Off
Trace Weighted Images	On
Tensor	Off
FA Maps	Off
ADC Maps	On
Exponential ADC Maps	Off
b-value >=	0 s/mm²
ADC Noise Threshold	100
Noise Masking	Off
Calculated Image	Off

Sequence - Part 1

Sequence Name	resolve
Dimension	2D
RF Pulse Type	Normal
Gradient Mode	Fast
Bandwidth	723 Hz/Px
Echo Spacing	0.32 ms
Optimization	Min. TE
EPI Factor	192

Sequence - Part 2

Introduction	On
Reacquisition Mode	On

Sequence - Assistant

SAR Assistant	Off
Optimization	Min. TE

\\Research\Investigators\Frederick\CORPUS_CONSISTENT\se

TA: 2:12 min Coil Selection: Auto Voxel Size: 1.0×1.0×5.0 mm³ Rel. SNR: 1.00

Properties

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slice Group	1
Slices	11
Distance Factor	100 %
Position	L0.0 A17.6 H8.8 mm
Orientation	Transversal
Phase Encoding Dir.	R >> L
Phase Oversampling	0 %
FOV Read	250 mm
FOV Phase	100.0 %
Slice Thickness	5.0 mm
TR	500.0 ms
TE 1	20.00 ms
TE 2	35.00 ms
Averages	1
Concatenations	1
AutoAlign	---

Contrast - Common

TR	500.0 ms
TE 1	20.00 ms
TE 2	35.00 ms
MTC	Off
Magn. Preparation	None
Flip Angle 1	90 deg
Flip Angle 2	180 deg
Fat-Water Contrast	Standard
Dark Blood	Off
Blood Suppression	Off
Contrasts	2
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Multiple Series	Off

Resolution - Common

FOV Read	250 mm
FOV Phase	100.0 %
Slice Thickness	5.0 mm
Base Resolution	256
Phase Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Deep Resolve	Off
Phase Partial Fourier	Off
Asymmetric Echo	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	2D
Normalize	Prescan
Image Filter	Off

Geometry - Common

Slice Group	1
Slices	11
Distance Factor	100 %
Position	L0.0 A17.6 H8.8 mm
Orientation	Transversal
Phase Encoding Dir.	R >> L
Phase Oversampling	0 %
FOV Read	250 mm
FOV Phase	100.0 %
Slice Thickness	5.0 mm
TR	500.0 ms
Multi-Slice Mode	Interleaved
Series	Interleaved
Concatenations	1

Geometry - AutoAlign

Slice Group	1
Position	L0.0 A17.6 H8.8 mm
Orientation	Transversal
Phase Encoding Dir.	R >> L
AutoAlign	---
Initial Position	L0.0 A17.6 H8.8
L	0.0 mm
A	17.6 mm
H	8.8 mm
Initial Orientation	Transversal
Initial Rotation	90.00 deg

Geometry - Saturation

Special Saturation	None

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	P >> A
Transversal	H >> F
Coil Combination	Adaptive Combine
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Tune up
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
A >> P	263 mm
R >> L	350 mm
F >> H	350 mm
Reset	Off

System - pTx

B1 Shim	TrueForm

System - Tx/Rx

Frequency 1H	123.249408 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	500.0 ms
Concatenations	1

Physio - Cardiac

Fat-Water Contrast	Standard
Magn. Preparation	None
Dark Blood	Off
FOV Read	250 mm
FOV Phase	100.0 %

Physio - Cardiac

Phase Resolution	100 %

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - MIP

MIP Sag	Off
MIP Cor	Off
MIP Tra	Off
MIP Time	Off
Radial MIP	Off
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Composing

Inline Composing	Off

Sequence - Part 1

Sequence Name	se
RF Pulse Type	Normal
Gradient Mode	Fast
Bandwidth 1	260 Hz/Px
Bandwidth 2	260 Hz/Px
Asymmetric Echo	Off

Sequence - Part 2

Introduction	On
Acoustic noise reduction	Off

Sequence - Assistant

SAR Assistant	Off
Allowed Delay	0 s

\\Research\Investigators\Frederick\CORPUS_CONSISTENT\csi_fid
TA: 36:24 min Coil Selection: Auto Voxel Size: 50.0×50.0×36.0 mm³ Rel. SNR: 1.00

Properties

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Position	L30.0 A10.7 H24.3 mm
Orientation	T > C-17.0
Rotation	0.00 deg
FOV A >> P	400 mm
FOV R >> L	400 mm
Thickness F >> H	36 mm
Vol A >> P	400 mm
Vol R >> L	400 mm
TR	3000.0 ms
TE	2.30 ms
Averages	60

Contrast - Common

TR	3000.0 ms
TE	2.30 ms
Flip Angle	90 deg
Preparation Scans	8
Averages	60
Water Suppression	None

Resolution - Common

FOV A >> P	400 mm
FOV R >> L	400 mm
Thickness F >> H	36 mm
Scan Res. A >> P	8
Scan Res. R >> L	8
Interpol. Res. A >> P	8
Interpol. Res. R >> L	8
Hamming	Off
Dimension	2D
Phase Encoding	Weighted
Vector Size	512
Normalize	Off

Geometry - Common

Position	L30.0 A10.7 H24.3 mm
Orientation	T > C-17.0
Rotation	0.00 deg
FOV A >> P	400 mm
FOV R >> L	400 mm
Thickness F >> H	36 mm
Vol A >> P	400 mm
Vol R >> L	400 mm

Geometry - AutoAlign

Slice Group	1
Position	L30.0 A10.7 H24.3 mm
Orientation	T > C-17.0
Phase Encoding Dir.	A >> P
AutoAlign	---
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Tune up
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off
Adj. Water Suppr.	Off

System - Adjust Volume

Position	L30.0 A10.7 H24.3 mm
Orientation	T > C-17.0
Rotation	0.00 deg
A >> P	400 mm
R >> L	400 mm
F >> H	36 mm
Reset	Off

System - pTx

B1 Shim	TrueForm

System - Tx/Rx

Frequency 31P	49.891876 MHz
Frequency 1H	123.249408 MHz
? Ref. Amplitude 31P	0.000 V
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	3000.0 ms

Sequence - Common

Sequence Name	csi_fid
Preparation Scans	8
Delta Frequency	0.00 ppm
Measurements	1
Dimension	2D
Save Uncombined	Off
Acquisition Delay	2.30 ms
Bandwidth	4000 Hz
Acquisition Duration	128 ms
Remove Oversampling	On

Sequence - Nuclei

TX/RX Nucleus	31P
TX/RX Delta Frequency	0 Hz
NOE Type	Rectangular
NOE Count	1
NOE Duration	5.00 ms
NOE Flip Angle	90.00 deg
TX Nucleus	1H
TX Delta Frequency	0 Hz
Decoupling Type	Waltz4
Decoupling Duration	1.00 ms
DC Total Duration	50.00 %
DC Pause Fract.	100.00 %
Decoupling Flip Angle	180.00 deg